Sustainable Operations and Computers 6 (2025) 198–216

Contents lists available at ScienceDirect

Sustainable Operations and Computers

journal homepage: http://www.keaipublishing.com/en/journals/sustainable-operations-and-computers/

The role of quality governance in achieving sustainable development goals in North Africa: An integrated decision-support system ☆

Muhammad Ikram*, Chaymae Boudraa

School of Business Administration, Al Akhawayn University in Ifrane, Avenue Hassan II, P.O. Box 104, 53 000 Ifrane, Morocco

A R T I C L E I N F O

Keywords:
Sustainable development
World governance
Sustainable development goals
North Africa
Integrated decision support framework

A B S T R A C T

North Africa is one of the most vulnerable regions to the impacts of climate change. Achieving the UN Sustainable Development Goals (SDGs) in North Africa requires effective governance, yet the governance factors influencing SDG progress remain underexplored. Therefore, this study examines the nexus between the Worldwide Governance Indicators (WGIs) of voice and accountability, political stability and violence, government effectiveness, regulatory quality, rule of law, and control of corruption and Sustainable Development Goals (SDGs) in North Africa. Previous research has insufficiently explored the relationship between global governance progress and the 2030 agenda. To bridge this gap, we applied an integrated Grey Relational Analysis (GRA) model to compute the ranks and weights the performance of North African countries in achieving the 17 SDGs from 2016–2022. The results show that Quality Education (SDG 4), Industry, Innovation and Infrastructure (SDG 9), Reduce Inequalities (SDG10), Sustainable Cities and Communities (SDG11), Responsible Consumption and Production (SDG12) and Peace Justice and Strong Institutions (SDG16) exhibited the highest performance in the region based on Second Synthetic Grey Relational Analysis Model (SSGRA). Additionally, a conservative maximin model was used to identify which North African country has the greatest impact of WGI on SDG outcomes. The findings indicate that Morocco faced the most significant challenges, particularly in achieving SDG 6 (clean water and sanitation). These results highlight critical implications for North African governments, emphasizing the need to prioritize (clean water and sanitation) SDG 6 and implement effective national strategies. This study recommends policymakers focus on governance reforms in lagging sectors and leverage data-driven frameworks to monitor and guide sustainable development interventions.

1. Introduction

Currently, the notion of sustainable development has emerged gradually throughout the world as an approach to address relevant challenges facing all societies. To address complex issues worldwide, including climate change, water sanitation, poverty and clean energy, the United Nations established 17 Sustainable Development Goals (SDGs) as urgent calls for action by all nations in 2015 (UN, 2015; [1]). To accomplish these goals, it is necessary to have ethical governance in which citizens are engaged and responsible for their own actions while adhering to the governance. Africa has recognized the crucial role of the SDGs in fostering a viable future for individuals and has taken part in countries that have pledged to work toward achieving these goals to achieve the global development agenda set for 2030 [2].

Sustainable development in the Middle East and North Africa (MENA) region is significantly lagging, with a 2023 regional assessment scoring MENA's SDG progress at only 59.8 out of 100, and 14 Arab countries failing to achieve a single SDG target [3]. This shortfall underscores an urgent practical need to understand drivers of SDG progress. Effective governance is widely recognized as a critical enabler of sustainable development, as governance quality influences economic, social, and environmental outcomes [4]. Academically, however, the link between governance and the full spectrum of SDGs remains underexplored, especially in North Africa. Most existing studies either focus on selective goals or broader regions for example, cross-country analyses have shown that governance improvements can reduce poverty and inequality, but these works did not cover all 17 SDGs nor focus on North African nations specifically. This leaves a gap in understanding how governance quality (across multiple dimensions) holistically affects sustainable development outcomes in this region. To address this gap, we ask: What is the impact of governance quality on the achievement of all 17 SDGs in North African countries?

However, a number of factors, namely, the rule of law and the level to which citizens are able to communicate their concerns and make those

☆ Peer review under the responsibility of Editorial Board of Sustainable Operations and Computers.

* Corresponding author.
E-mail address: i.muhammad@aui.ma (M. Ikram).

https://doi.org/10.1016/j.susoc.2025.07.005
Received 28 October 2024; Received in revised form 12 April 2025; Accepted 31 July 2025
Available online 8 August 2025

in charge accountable, influence the way SDGs are carried out. According to World Bank, [5], the rule of law can be defined as the extent to which a country's citizens trust their policymakers and adhere to the regulations and laws of their society. According to the World Bank, the rule of law covers several sub-indicators measuring the perception level of citizens toward legal standards. This consists of thoughts regarding the frequency of crimes, the reliability of the judiciary and the execution of contracts. Overall, these parameters evaluate whether a nation has developed a framework for a relationship between agents and the economy that relies on consistent laws and, significantly, how property rights are effectively protected.

Institutional inadequacies and the enduring legacy of corruption represent two major obstacles to attaining the SDGs [6]. In many developing nations, corruption significantly diminishes the efficacy of governmental institutions, impairing their ability to foster social and environmental welfare [7]. This institutional inefficacy impedes important stakeholders from executing their missions, ultimately hindering progress toward the SDGs. Furthermore, the deleterious impact of corruption obstructs economic progress, suppresses innovation, and leads to considerable economic losses [8], hence further hindering sustainable development initiatives in these areas.

Voice & accountability is an umbrella term that covers a wide range of rights to be secured worldwide [8]. It embraces the perceptions of the degree to which a nation's citizens hold total freedom of expression, the ability to choose a preferred governing party, the right to promote common rights and the existence of media that operates freely without any restrictions [9]. In this response, local authorities that can be held responsible for their acts are inclined to satisfy citizens' requests and collective desires. In other words, "Voice" and "Accountability" are considered two complementary terms. The World Bank states that Voice & Accountability is one of the six essential characteristics of governance encompassed in the WGI, which also encompass government effectiveness, political stability and absence of violence, regulatory quality, control of corruption, and rule of law [10].

Political stability is essential for attaining long-term SDGs in rising economies since it aids in reducing CO2 emissions, given that political risk encompasses both internal and external conflicts inside a nation [11]. Therefore, it is crucial to analyze the impact of political stability on CO2 emissions in conjunction with other contributing elements, such as renewable energy consumption and green finance, to guarantee advancement toward long-term SDG targets.

Governance challenges in North Africa are significant and alarming. The 2015 worldwide governance report World Bank, (2015) positioned countries in this area among the lowest in all six governance criteria. In 2015, Sudan was classified in the 178th percentile, whereas Morocco, Tunisia, Algeria, and Egypt were positioned 100th, 108th, 148th, and 135th, respectively. Conversely, Fig. 1 depicts the trends of multiple governance indices for North Africa from 2016 to 2022.

The United Nations SDGs consist of a framework of 17 interrelated objectives aimed at jointly addressing the three fundamental pillars of sustainability: economic, social, and environmental. These goals provide a comprehensive agenda aimed at fostering global progress in achieving long-term sustainable development across multiple dimensions [1]. The United Nations approved the SDGs, also known as global goals, in 2015, and they are planned to be completed by 2030 under an international development framework. At the UN summit, the SDGs are called for worldwide to address poverty, protect the environment, and promote social welfare [12]. This is the framework for a progressive and sustainable future, which is consistent with the commitment of all nations to take appropriate action. The 17 goals are formulated cohesively, since they are interrelated in that the advancement of one goal facilitates the growth of another. Progress toward the SDGs varies across Africa's subregions. West Africa and North Africa show the strongest performance, whereas East Africa lags behind. However, no subregion is currently on track to achieve the SDGs fully. A significant challenge in Africa is the lack of comprehensive data to monitor progress accurately [13]. Additionally, lockdown measures, supply and demand disruptions, and interruptions in global value chains have severely impacted progress. Economically, these challenges have resulted in decreased export revenues, tourism, and remittances and reduced capital flows while exacerbating social issues such as inequality, poverty, hunger, and unemployment [14].

Algeria, Tunisia, and Morocco lead the region with average index scores exceeding 69 (on a scale of 0–100), ranking highest in the continent and subregion. In terms of the global ranking in the 2024 report, Tunisia (60) and Morocco (69) are well positioned given their income levels among the 163 countries, followed by Algeria (71) and Egypt (83), which hold middle-tier rankings, whereas Sudan (159) ranks near the bottom [15].

In addition to the lingering effects of COVID-19, the Russia–Ukraine war has severely affected North Africa, particularly through rising food, fuel, and fertilizer prices, as well as supply chain volatility. The current global context, with ongoing challenges in energy and food security, has slowed progress toward achieving the SDGs in North Africa. This underscores the urgent need for governments to accelerate reforms and initiatives aimed at sustainable development. Fig. 2 presents the SDG progress in North Africa from 2016 to 2024.

Despite growing literature on governance and development, no prior study to best of author's knowledge has analyzed all 17 SDGs in the specific context of North Africa using an integrated framework. Most related studies have been either global or focused on individual aspects. For example, Deb [16] examined governance impacts on poverty and inequality in a cross-regional sample, and Adebayo et al. [17] found that overall governance quality strongly correlates with broad sustainable development indicators in Africa. Recent advancements in rural-focused augmented reality (AR) technologies, integrated within platforms such as TikTok and Kuaishou, have significantly enhanced farmers' ecological cognition, suggesting a strong potential for improvements in technology-driven governance. A recent widespread power outage in northern Chinese provinces highlighted the importance for decentralized renewable energy infrastructure, emphasizing the role of effective governance in achieving energy resilience. Initiatives utilizing drones for social media outreach in mountainous regions with restricted internet connectivity have significantly enhanced ecological awareness and supported governance efforts aimed at sustainable energy transitions [18]. Additionally, recent research has emphasized that rural energy poverty significantly restricts the ability of communities to transition effectively to clean energy, underscoring the critical role governance plays in facilitating energy equity. This scenario becomes even more pressing given China's ambitious targets to achieve carbon neutrality by 2060, highlighting the necessity for decentralized energy governance systems that align sustainability goals with local resource availability and community needs [19].

This study highlights that North African nations are at a critical juncture many are off-track on the SDGs with only a few years left to 2030, and governance reforms are intensifying in the post-Arab Spring era. This study contributes in many ways: we integrate all SDGs and a full set of governance indicators using a novel Grey Relational Analysis approach, whereas previous works looked at subsets of goals or used traditional regression methods. In terms of findings, this study anticipate alignment with prior studies (e.g., that better governance facilitates development, as others have suggested), but by drilling down to specific SDGs we also uncover new insights (for instance, pinpointing which SDGs are most sensitive to governance deficits in North Africa). These recent developments reinforce the importance of our integrated approach now.

Moreover, this study proposed grey relation analysis (GRA), a technique particularly suited to this study's objectives, we obtained more persuasive and reliable findings while mitigating endogeneity issues related to variable acquisition. The GRA methodology's primary advantage is its capacity to yield precise results despite constrained datasets, rendering it particularly suitable for evaluating SDG achievement in North African nations with potential data deficiencies. Moreover, GRA

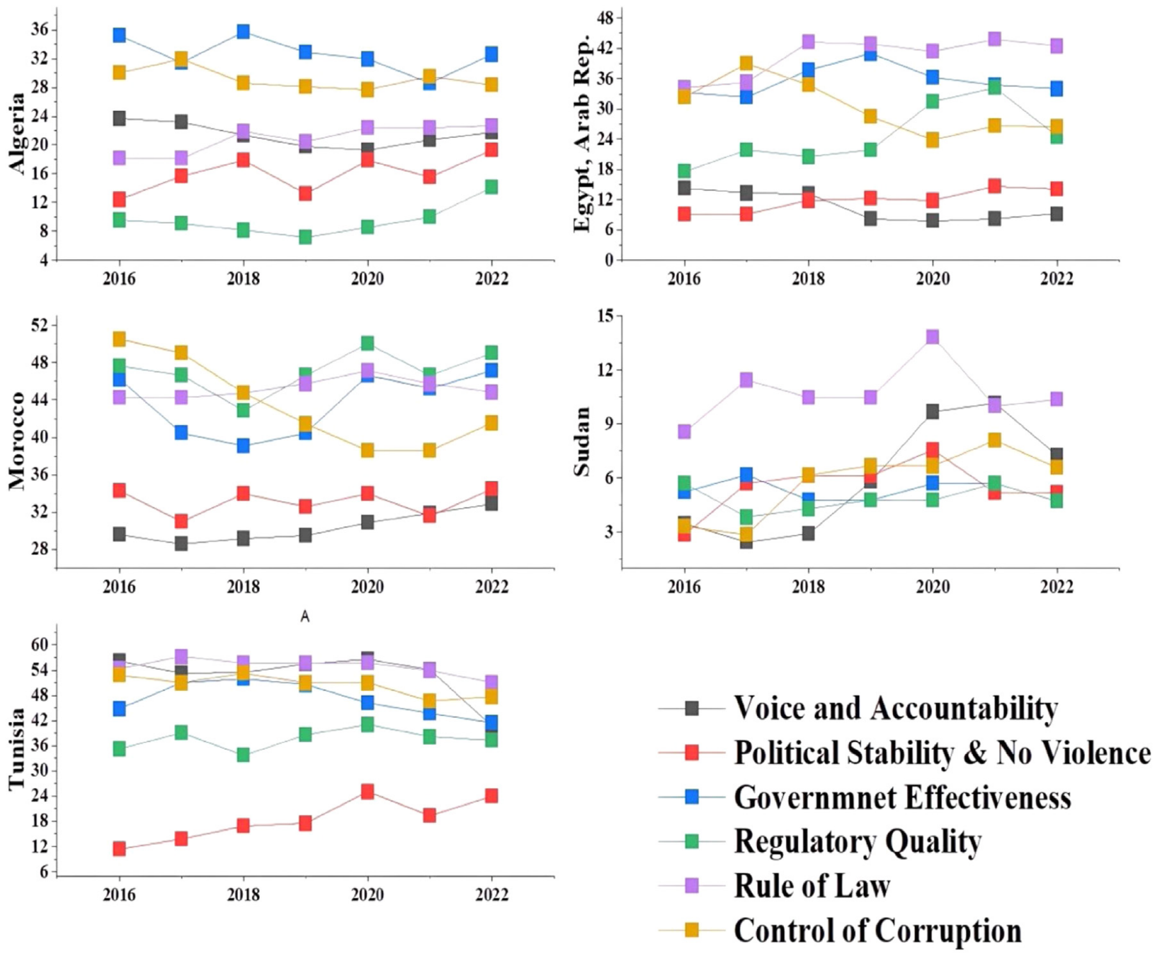

Fig. 1. Governance performance of North Africa from 2016–2022.

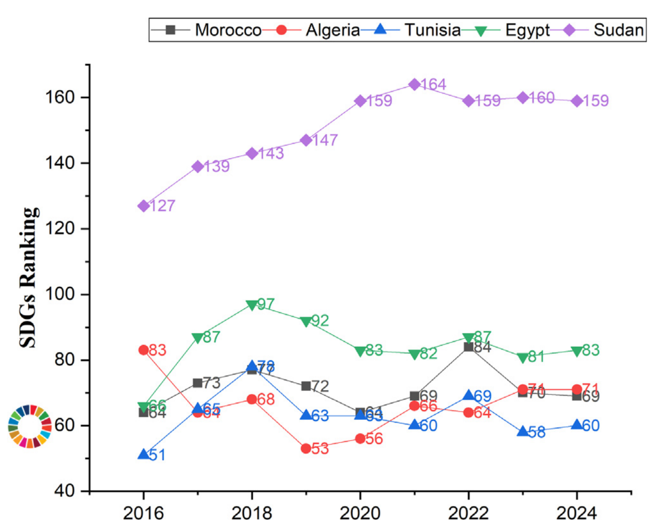

Fig. 2. Sustainable development goal (SDG) progress in North Africa.

models are capable of managing absent data without undermining the validity of the analysis. Additionally, the conservative (maximin) criteria method was implemented in this study, which enables a comprehensive examination of the average impact of governance indicators on SDG performance. This approach enhances the robustness of the findings by focusing on minimizing the worst-case scenarios, providing a clearer understanding of how governance factors influence sustainable development.

This study aims to examine the nexus between WGI and the SDGs, identifying the socioeconomic, demographic, and political barriers that have delayed progress toward the 2030 Agenda. A key objective is to pinpoint the SDGs most affected in Morocco, Algeria, Tunisia, Sudan, and Egypt, enabling national governments to make informed, targeted decisions. By filling a critical gap in the literature, this research offers actionable insights for policymakers and decision-makers, providing them with a clear understanding of how governance indicators influence SDG outcomes. It establishes a robust framework for monitoring and evaluating SDG progress in the region, ultimately empowering North African governments to take decisive action and implement effective strategies for achieving the 2030 Agenda.

The paper is organized into five sections. The second section reviews the literature and examines how the SDGs are related to WGI in North Africa. The research technique is described in Section 3 and comprises the conservative maximin model, second synthetic grey relational analysis (SSGRA), and grey relational analysis. The results are presented in section four, along with a thorough analysis of the data. The conclusion is presented in the last part, which also highlights the study's research limitations and future directions.

2. Literature review

2.1. Governance and sustainable development

Governance is the ability of a government to create, carry out, and uphold laws that have an impact on the people and institutions that make up a nation. It can be defined by governmental institutions that are largely influenced by the acts of federal agencies, especially at the

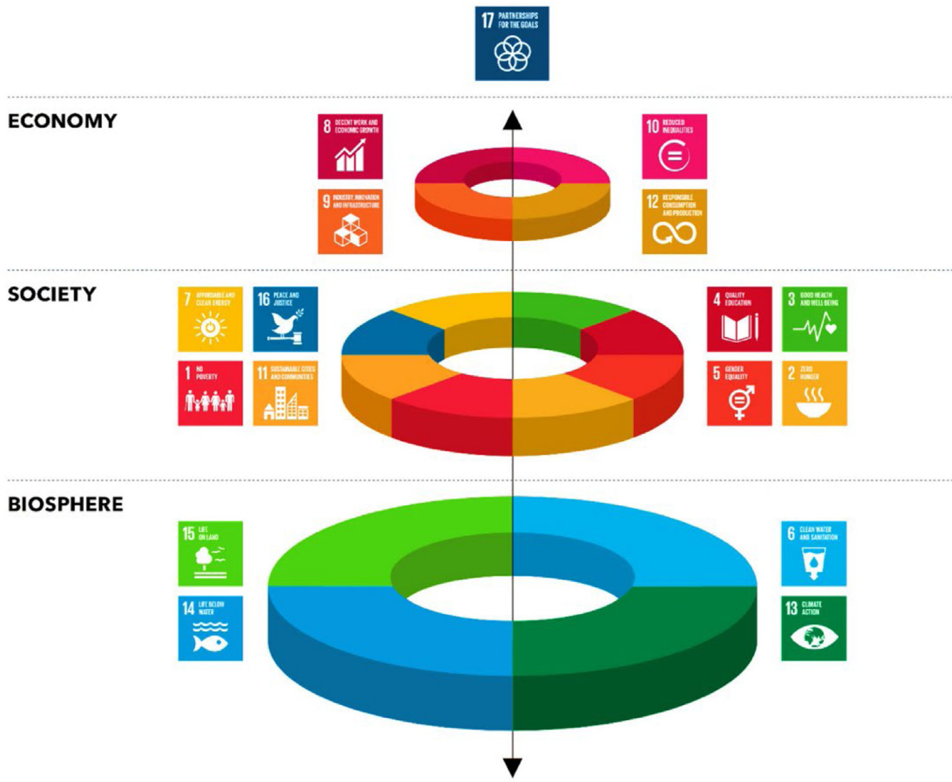

Fig. 3. UN Sustainable Development Goals (Source: [25]).

national level. It is frequently considered a customary framework inside society [20]. Six essential components of good governance were recognized by [21]: government effectiveness, political stability and the absence of violence, regulatory quality, the control of corruption, and the rule of law. Furthermore, Kaufmann et al. [21] emphasized that various nations have varied levels of effect from governance indicators on macrolevel results.

The importance of good governance in expediting the attainment of the SDGs is underscored by scientific investigations about sustainable development. The rule of law, which promotes accountability and transparency in governmental operations, is a crucial sign of moral governance [22]. Prior studies have indicated that communities possessing a robust legal system typically outperform others in terms of a range of development objectives. In the same way, accountability and voice are crucial to achieving the SDGs [23,24]. Public needs and national policies are better aligned when citizens actively participate in decision-making processes and hold leaders accountable for their actions. The results of development are greatly aided by this vigorous civic engagement.

To obtain a good understanding of each SDG, it is critical to know the main objectives of each goal separately and enable their analysis with respect to the rule of law and their connection with voice and accountability in a clear manner. According to the Sustainable Development Goals report, the 17 SDGs can be categorized into three different sets: economy, society and environment (2022), as presented in Fig. 3.

Moreover, unemployment is exceedingly common, particularly in the aftermath of the worldwide pandemic, indicating that it is even more crucial to foster resilient economic growth, together with ethical labor conditions [26]. The goal is to encourage innovative inventions, scientific research, and technology development, primarily in developing nations. The goal is to achieve equal opportunity by eradicating discriminatory legislation and regulation. Responsible consumption and production (SDG12) patterns must be put into action along with the adoption of the UN's ten-year framework of programs. This final objective strives to support the collaboration agreement and commitment of all the nations in the SDGs.

The aim is to encourage the implementation of protective measures such that everyone has equal rights to financial resources and vital necessities: food, shelter, and clothing. This approach eradicates hunger and establishes a sustainable agricultural framework that mitigates the environmental impacts associated with increased food production [27]. The intention is to ensure that every child, teenager, or adult has access to education and training programs (SDG4: good health & well-being) that are suitable for their requirements. Additionally, equity and gender equality in education are important (SDG5: gender equality). This goal also fights against discrimination and all types of violence that threaten women [28]. The goal (SDG 7: Clean and affordable energy) promotes the availability of reliable energy resources to all people. It also encourages investment in energy sustainability [29]. This aim advocates the adoption of national urban policies, increased accessibility to public areas and easy access to public transit. The intent is to advance a peaceful world and promote a reduction in all sorts of violence.

2.2. Sustainable development in North Africa

In 2022, the sustainable development index of Morocco has dramatically declined, which resulted in a double-digit fall in its overall ranking, shifting from 69th to 84th out of 165 nations. Although Morocco earned 69 out of 100 in the SDG report, it is still considered above the MENA average, which represents a score of 66.7, placing Morocco third in North Africa (Morocco World News, 2022). Through initiating action on climate change alongside ethical consumption and production, Morocco has preserved two SDGs. Note that the Moroccan government is on schedule to provide access to clean water and sanitary facilities for all of its residents, regardless of the substantial difficulties in tackling this matter. However, extreme weather events and very little rainfall slow economic activities since agriculture is the primary economic sector that contributes 14 % of the country's GDP and >80 % of the country's water resources [30]. Currently, the country is involving civil society to adopt the necessary mechanisms for addressing water scarcity and improving sustainable measures.

Egypt has developed innovative drip irrigation systems to minimize water consumption and is actively working to implement the SDGs. This led to a 40 % reduction in the level of fresh water consumed for agriculture in 2021 and 2022. Drip irrigation technology can help Egypt's agricultural industry meet six major goals: SDGs 1, 2, 3, 6, 8, and 11. Egypt has also incorporated information technology to advance SDG progress through broad data gathering and analysis using computational tools to identify environmental patterns, individual behavior, and experiences that assist policy makers in evaluating performance [31].

However, the government of Algeria has recently placed more emphasis on three core legislative priorities: industry, innovation, and trade. The selection of these particular criteria is supported by Algeria's current national roadmap toward achieving sustainable consumption and production modes for 2030 [32]. Algeria is putting effort to help further sustainable development by ensuring that this aspect is incorporated into all official and corporate strategies and programs, as well as the initiatives of civil society and individual citizens. Moreover, creating job opportunities and improving quality of life are both priorities for Algeria; therefore, the country is doubling in terms of innovation [33].

In the midst of the current political conflict in Tunisia, organizational and political instability has a significant effect on the SDG reporting of oil firms [34]. Companies' responses to political interference and shifts are also likely to be influenced by the political environment in which they operate. Firms appeared reluctant to fund SDG initiatives because of concerns about the reliability of formal institutions and political regimes. This, in turn, creates a context that is not conducive to maintaining SDG records, which might constrain and even impede the implementation of SDG reporting procedures in developing countries [35].

Hussain et al. [36] investigated how the rule of law and government size influence the efficiency of social and financial microfinance institutions (MFIs), confirming that the rule of law in a nation significantly affects economic development. Moreover, the rule of law, property rights and government integrity had strong positive relationships with microfinance institutions' financial efficiency. Thus, the rule of law contributes to the achievement of SDG 8 economic growth. According to Arajärvi, [37], the rule of law is one of the criteria adopted to evaluate the quality of public services, as it is one of the factors representing good governance. Within an environmental context, the rule of law plays an important role in fostering a sustainable environment [38]. The concept of the rule of law aligns with the elements of constitutionalism, claiming that governmental leaders, policy makers, citizens, and entities are expected to adhere to the same norms of laws related to the protection of human rights. These regulations restrict nations from urbanization, reducing the consumption of CO_2 emissions and employing more renewable resources for the sake of human welfare (Li, 2023). According to Miliute-Plepiene et al. [39], in the Nordic region, authorities' initiatives to guarantee long-term sustainability revealed that the governance of Nordic countries considers the rule of law principles to increase GDP per capita.

In Morocco, the overall rule of law index for 2023 increased by four digits compared with the previous year, with a total ranking of 92 out of 142 nations worldwide, whereas Denmark takes the lead within the list with the first top rank (World Justice Project, 2023). Whereas the underlying factor, such as government power restrictions, Morocco held the position of 76, with a score of 0.51. Additionally, with respect to the absence of corruption, Morocco's global rank was 101, and at the regional level, it was 3 out of 9 countries. With respect to fundamental rights, the overall ranking was 114 out of 142, the regional ranking was 7 out of 9, and Morocco secured an average score of 0.43, whereas the regional average was 0.40. The country secured a spot of 66 on the regulatory enforcement factor across the globe, and the regional rank declined, with one position from the previous year, to represent 4 out of 9 in 2023, resulting in an average score of 0.50.

With the aim of meeting the accountability related to the SDGs by 2030, North Africa. In recent years, Moroccans have been able to voice their concerns on many occasions. For example, in 2018, there was a widespread citizens boycott of consumed goods perceived to be overly expensive and associated with corporate monopolies [40].

Climate change is one of the most significant challenges that the world is currently facing with, and it has a detrimental impact on the efforts of nations to achieve sustainable development [41]. This is especially true for North Africa, which has disproportionate consequences despite having a small contribution to global emissions. Compared with other regions of the world, the region is expected to suffer greater economic losses [42]. An increase in the frequency of extreme weather events such as heatwaves, droughts, and floods is one of the first signs of climate change in North Africa. Other symptoms of climate change include rising global temperatures, rising sea levels, and acidification of the ocean. Threats to food security, diminishing water supplies, decreased productivity of natural resources, biodiversity loss, deteriorating public health, increased land degradation, desertification, and coastline erosion are only a few of the severe effects of these occurrences [43].

North African countries are making progress toward several SDGs, including SDG 17 (Partnerships for the Goals), SDG 16 (Peace, Justice, and Strong Institutions), SDG 13 (Climate Action), SDG 12 (Responsible Consumption and Production), SDG 9 (Industry, Innovation, and Infrastructure), SDG 6 (Clean Water and Sanitation), and SDG 4 (Quality Education). However, significant challenges persist in addressing SDG 1 (no poverty), SDG 5 (gender equality), SDG 8 (decent work and economic growth), and SDG 10 (reduced inequality) [44]. The detailed progress of the 17 SDGs of North African countries is presented in Fig. 4.

2.3. Integrated governance-SDGs framework

This study selects the governance factors and SDG outcomes is grounded in development theory and prior empirical findings. Governance theory (e.g., institutional theory) posits that stronger institutions and governance practices create an enabling environment for development [45]. Consistent with this, studies have found that improvements in governance such as reduced corruption, increased political stability, and better regulatory quality tend to positively affect human development indicators. For instance, recent evidence suggests governance quality is critical for poverty alleviation and equality

Based on the body of literature, we focus on the WGI as our independent variables because it captures key institutional dimensions (voice and accountability, political stability, government effectiveness, regulatory quality, rule of law, control of corruption) that are theoretically linked to development outcomes. Likewise, the SDGs represent a comprehensive set of development targets (economic, social, and environmental) aligning with the triple bottom line of sustainable development theory. By examining the association between WGI and all SDG indicators, our framework extends prior studies that often looked at narrower pairings (e.g., governance vs. GDP or single social outcomes). We acknowledge that similar relationships have been studied in literature. Unlike conventional methods such as regression-based analyses found in the literature, we employ a multi-criteria decision method GRA to capture the strength of association between governance and each SDG in a comparative manner. This allows us to analyze patterns and rankings of influence in a way that previous studies have not.

Building on the theoretical and empirical rationale underlying the selection of WGI and SDG indicators, the conceptual framework depicted in Fig. 5 illustrates how governance quality, captured by the WGI, influences SDGs performance through a country's socioeconomic, demographic, and political challenges. In this framework, SDG performance is explicitly categorized into economic, societal, and environmental domains, aligning with the triple-bottom-line approach and underscoring that sustainable development spans all three pillars. Furthermore, the model posits that high-quality governance acts as a structural enabler, creating a conducive environment for more effective management of socioeconomic hurdles, demographic pressures, and political instabilities, thereby fostering improved outcomes across all SDG domains. At

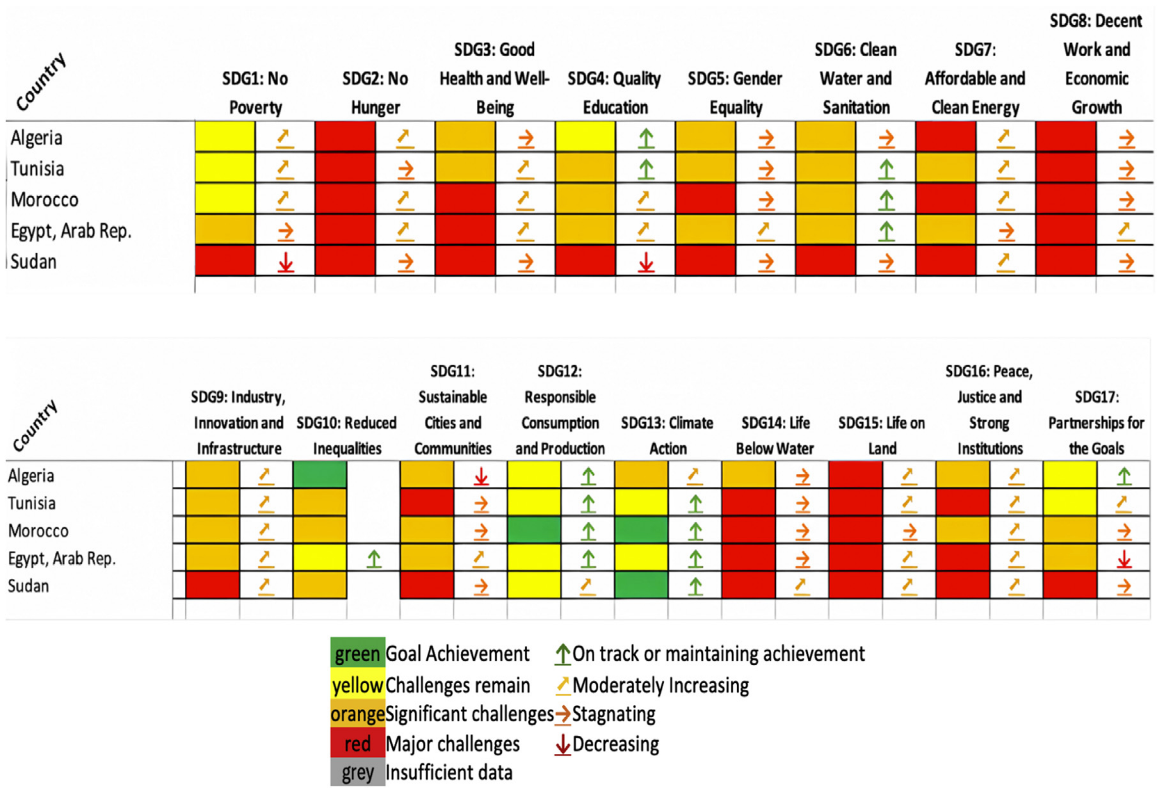

Fig. 4. 17-SDGs progress in North African countries (UN, 2023).

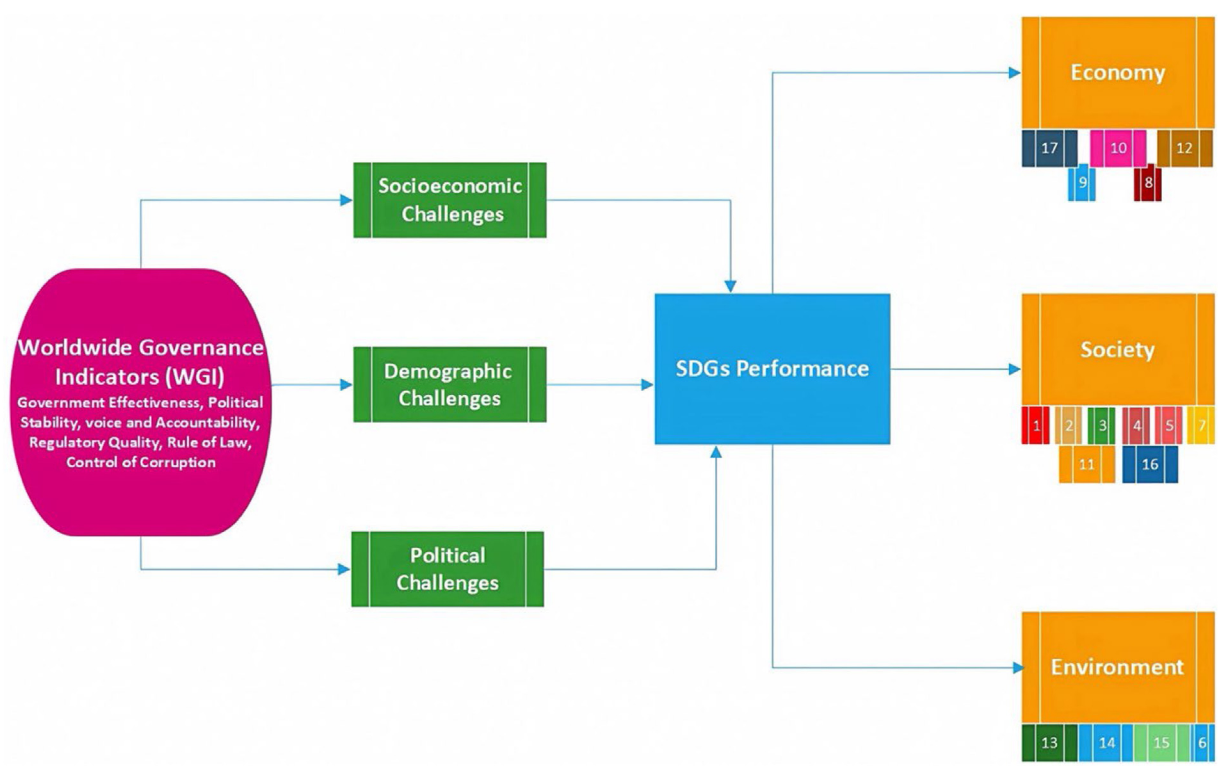

Fig. 5. Research framework of the study.

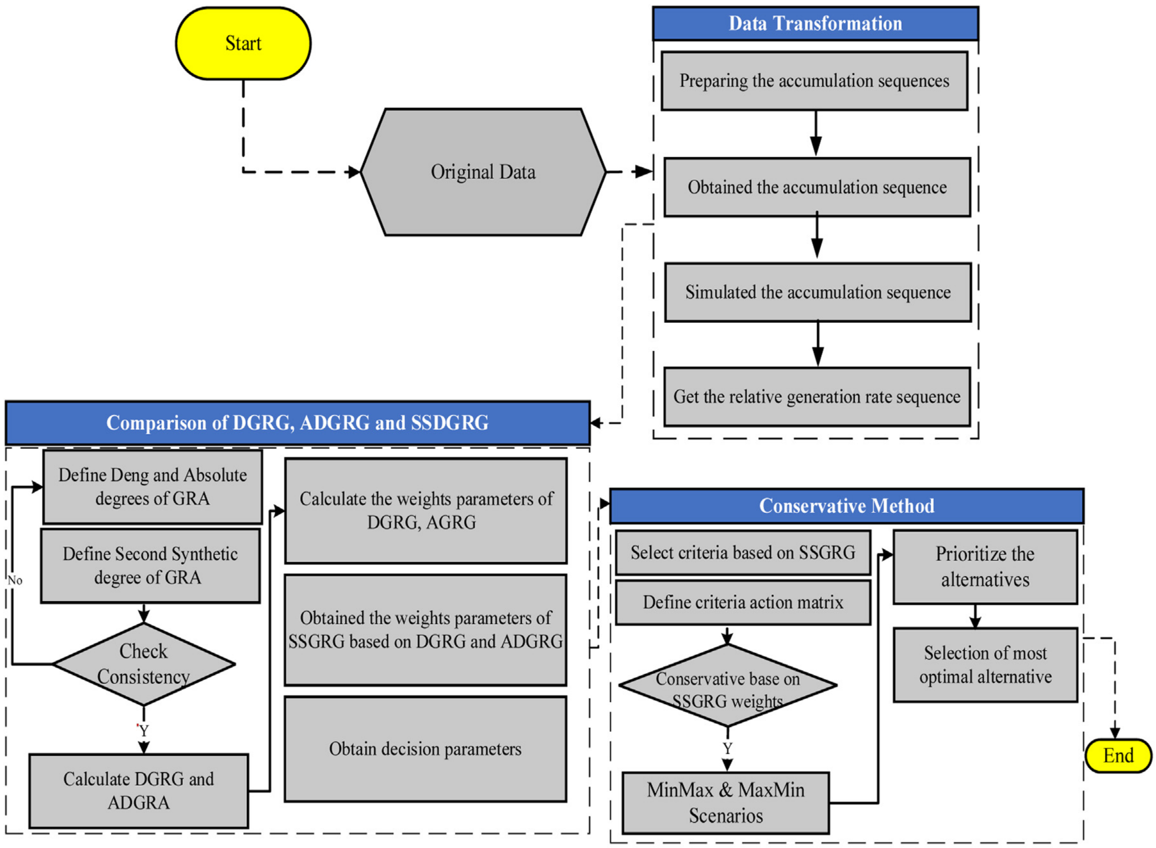

Fig. 6. GRA-based methodological framework operationalized in this study.

the same time, governance is operationalized as a moderating factor in the framework, shaping development trajectories by conditioning how strongly these national challenges affect outcomes: robust institutions can mitigate adverse impacts, whereas weak governance may exacerbate them.

3. Research and data methodology

In this section, we detail the step-by-step procedure of our analysis. Fig. 6 provides an overview of our integrated decision-making framework linking governance indicators to SDG outcomes. We first compile the dataset of governance metrics (WGI) and SDG performance indicators for each country (Section 3.1). We then apply the GRA technique to evaluate the strength of association between each governance dimension and each SDG (Section 3.2). GRA is well-suited for this study as it is a multi-criteria decision-making method that can handle complex, small-sample comparative evaluations; its advantages include using original data directly and yielding easy-to-interpret relational grades.

We adopted GRA in this study because it offers several benefits aligned with our research needs. GRA is designed to analyze systems with uncertain or incomplete information, making it ideal for socio-economic comparisons where data may be limited, or indicators are measured in different units. It allows us to derive a quantitative measure of the strength of association (the grey relational grade) between each governance indicator and each SDG outcome. Importantly, GRA has been successfully used as a decision-making tool in various multi-criteria contexts, demonstrating its reliability for ranking and evaluating alternatives [46].

In our scenario, alternative profiles (each country's governance-SDG performance) are compared across multiple criteria consistently. We considered other methods (such as regression analysis or composite indexing), but those would either not capture the pairwise goal-specific relationships we are interested in or require larger sample sizes for robust statistical inference. GRA, by contrast, is well-suited for a small sample of countries and focuses on relational comparison rather than inference, which aligns with our exploratory objective. Therefore, we chose GRA to leverage these strengths and to gain nuanced insights into how each governance factor relates to each development goal.

Our methodology follows established practices in GRA [47,48] and is informed by similar applications in sustainability research. For example, grey based multi-criteria approach has been used to assess technology adoption factors in rural sustainability contexts. Yin et al. [18] extended the UTAUT2 model to analyze clean energy adoption drivers, illustrating the value of a structured model-building approach. By clearly explaining our model-building process and citing analogous methodologies from prior studies, we ensure the rigor and transparency of our research design.

3.1. Data

This research aims to examine how WGI contributes to the 17 SDGs in North African countries, including Algeria, Sudan, Egypt, Tunisia, and Morocco. Governance performance is evaluated via the established framework of Kaufmann et al. [21], which provides a comprehensive set of indicators to assess the effectiveness of governance in these nations. The six governance indicators were employed: government effectiveness (GEF), political stability (PLS), regulatory quality (RQA), control of corruption (COC)., and rule of law (RLW). Seventeen sets of SDG data were retrieved from 2016–2022 from the SDG transformation center for North Africa. MATLAB analysis was performed after all variables were normalized to a range of 0–1. With data from the UN SDGs, World Bank, [49], the study concentrated on the years 2016–2022. An overview of

Table 1
Variable descriptions, units for measurement and data sources.

Variables (codes)	Description of variables	Measurement unit	Data sources
SDGs	Sustainable Development Goals	17 Sustainable Development Goals scores	UN SDGs
VAA	Voice and accountability	voice and accountability estimate value	WGI
COC	Control of corruption	Control of corruption estimates value	
PLS	Political stability	Political stability estimates value	
RQA	Regulatory quality	Regulatory quality estimates value	
GEF	Government effectiveness	Government effectiveness estimates value	
RLW	Rule of law	Rule of law estimate value	

the variables, their means of measurement, and the sources of the data is given in Table 1.

3.2. Grey relational analysis (GRA) models

Grey Relation Analysis (GRA) systems are considered an early phase in the conceptual development of grey models. The foundation of grey models was established by Julong Deng in 1982. Initially, the applications of GRA has been applied across various fields to address limitations arising from incomplete data and unclear mechanisms. One of the key advantages of GRA is its practical application even with limited data samples [50]. In 1985, Deng introduced the GRA system and proposed the idea of grey relational grade (GRG), which improved the methodology's ability to analyze complicated systems with a small set of information.

Rehman et al. [51] reported that the grey model has a significantly greater degree of accuracy than statistical models do. The advantage of the grey method is ensuring the timeliness of data alongside its accuracy during the analysis phase. The primary principle of GRA models aims to assess and quantify correlations by utilizing the closeness of a geometric curve across a small dataset (Liu et al., 2017; p 68) [52].

The GRA system is composed of three distinct categories of closeness. The absolute GRA measures the overall proximity between two data sequences, whereas the synthetic GRA seeks to capture a broader or more inclusive sense of closeness. Nonetheless, Deng's GRA hypothesis pertains just only to partial proximity (Ikram et al., 2022).

3.3. Deng's GRA model

Various scholars have proposed numerous GRA systems, but Deng's 1985 model remains the most widely recognized and relevant, making it crucial for understanding grey systems [53]. Deng's degree of GRA paradigm has been applied to a number of real-world issues in a variety of domains, whether in regard to conceptual or practical topics, taking into account the basic premise of this model [54]. The following is a list of the two data sequences that X_0 and X_1 presented for Deng's grey model prediction.

3.3.1. Deng degree of the GRA model

The initial outputs of X_0 and X_i, where $i = 1, 2,3\ldots, m$, are calculated in the first stage, where:

$$X'i = Xi/xi(1) = \left(x'i(1), x'i(2), \ldots, x'i(n)\right); i = 0, 1, 2, 3 \ldots, m \tag{1}$$

The distinction among X'_0 and X'_i and $i = 0, 1, 2, 3 \ldots, m$ was computed in the second phase as follows:

$$\Delta i(l) = x'0(l) - x'i(l), \Delta = (\Delta i(1), \Delta i(2), \ldots, \Delta i(n)), i = 1, 2, 3, \ldots, m \tag{2}$$

The third stage covers the greatest and minimum difference, which is expressed as follows:

$$M = \max_i \max_l \Delta_i(L), m = \min_i \min_l \Delta_i(L) \tag{3}$$

Moreover, in the fourth stage, the following formulas are used to create the grey incidence constants:

$$\gamma_{0i}(l) = \frac{m + \xi M}{\Delta i(l) + \xi M}; \xi \in (0, 1); l = 1, 2, 3 \ldots, n; i = 1, 2, 3 \ldots, m \tag{4}$$

The distinct coefficient is denoted by ξ and is taken to have a value of ξ=0.5. The fifth and final step entails projecting the grey incidence degree, also known as grey incidence Deng's degree, as follows:

$$\gamma_{0i} = \frac{1}{n}\sum_{l=1}^{n} \left(\gamma_{0i}(L)\right); i = 1, 2, 3 \ldots, m \tag{5}$$

The fractional representation of 1/n can be expressed using weights wl as follows:

$$\gamma 0i = L = 1/n(\gamma 0i(L) \times wl); i = 1, 2, 3 \ldots, m \tag{6}$$

If the system is affected differently by each factor, then the total weights will equal one, or $\sum wl = 1$. The explanation can be found in Sarkaya and Güllü (2015). This method suggests that $1/n$ infers that the weighting variables are spread equally, but w_k is a weighting method with an unequal distribution that is frequently used in real-world tasks [50].

3.4. Absolute degree of the GRA model

According to Silva et al. [54] the ADGRA, or absolute GRA model, helps to clarify the relationships among various factors, whereas the Deng GRA emphasizes the influence of GRA components represented within the grey database. By analyzing the geometric closeness of the succession curves, the Deng and absolute degrees of GRA models are developed to achieve the primary objective, proving highly valuable in research applications.

The method used to determine the exact grey relation between X_0 and X_1 over two identically distributed time periods is as follows:

In the very first stage, X_0^0 and X_1^0 zero initial values, denoted as X0 and X1, are established using the interval formula. The second stage entails the computation of $|s_0|$, $|s_1|$ & $|s_1 - s_0|$. The absolute degree of grey incidence (ε01) is forecasted across the discrete time pattern between X_0 and X_1. Consequently, the following formula is employed:

$$\varepsilon_{01} = \frac{1 + |s_0| + |s_1|}{1 + |s_0| + |s_1| + |s_1 - s_0|} \tag{7}$$

Deng's GRA model, known as Deng's grey relation analysis (DGRA) model, proposes the Deng degree of grey incidence or GRG. It is presented as:

$$\gamma_{0i} = \sum\nolimits_i^n \left(\gamma_{0i}(L) \times w_l\right); i = 1, 2, 3, \ldots, m \tag{8}$$

Or

$$\gamma_{0i} = \frac{1}{n}\sum_{l=1}^{n} \left(\gamma_{0i}(L)\right); i = 1, 2, 3, \ldots, m \tag{9}$$

The coefficient of grey incidence, also known as the grey relational coefficient, is expressed as γ0i (L). The DGRA model's first equation represents the weighted version of the GRA, and its second formula represents a nonweighting arrangement.

An absolute degree of grey relation (or ADGRA) was first proposed in the absolute GRA model. It is described as follows:

$$\varepsilon_{01} = \left(1 + |s_0| + |s_1|\right)/\left(1 + |s_0| + |s_1| + |s_0 - s_1|\right) \tag{10}$$

If we write the series of system behavior data as $X_i = (x_i(1), x_i(2), \ldots, x_i(n))$, which is the fulfilling sequence operator, and

$X_i(L)_c = (x_i(L)_c - x_i(1); L = 1, 2, 3, ..n$, then C is a zero-based point operator and X_iC is Xi's zero-scale point picture. Typically, X_iC may be abbreviated as $X_iC = X_1^0 = (x_i^0(1), x_i^0(2), \ldots \ldots x_i^0(n))$.

Huang et al. [55] suggested that X_i and X_j are zero-initial point sequences with equal lengths, representing a single time-interval arrangement factor.

$$X_i^0 = \left(x_i^0(1), x_i^0(2), x_i^0(3) \ldots, x_i^0(n)\right),$$

$$X_j^0 = \left(x_j^0(1), x_j^0(2), x_j^0(3) \ldots, x_j^0(n)\right) \tag{11}$$

Then,

$$|s_i| = \left|\sum_{i=2}^{n-1} x_i^0(L) + \frac{1}{2}x_i^0(n)\right|$$

and

$$\left|s_j\right| = \left|\sum_{i=2}^{n-1} x_j^0(L) + \frac{1}{2}x_j^0(n)\right| \tag{12}$$

The absolute distinction among $s_i - s_j$ is found via the following formula:

$$\left|s_i - s_j\right| = \left|\sum_{i=2}^{n-1}\left(x_i^0(L) - x_j^0(L)\right) + \frac{1}{2}\left(x_i^0(n) - x_j^0(n)\right)\right| \tag{13}$$

According to Javed et al., (2019), the second synthetic grey relational analysis (SSDGRA) model is defined as follows:

$$\rho_{ij} = \theta\varepsilon_{ij} + (1-\theta)\gamma_{ij}; \theta \in [0,1], \tag{14}$$

The absolute degree of grey relation (ε_{ij}), Deng's grey relational grade (GRG), and (γ_{ij}) are indicated, with ρij representing a secondary synthetic degree of grey incidence or grey relation. A key component of Deng's grey incidence model is the coefficient of grey relation or incidence at specific points. The absolute relational model is based on a comparatively broader perspective [56]. The SSDGRA plays a crucial role in establishing the overall relationship between two sequence levels.

The SSDGRA model was developed based on the operational standard of the SDGRA model. To attain a comprehensive positioning that equally incorporates the advantages of the two parameters, γ and ε, without bias, setting θ=0.5. If necessary, θ may also be altered. When someone wants to support anything, the calculation might be reduced, but when one desires something, the overall worth might also be increased. The methodological framework operationalized in this study is presented in Fig. 6.

4. Results

In this study, we investigate the relationship between WGI and SDGs by using the GRA models of Deng GRA, absolute GRA, and SSGRA. The interrelationship between the WGI and the SDGs is assessed on a 0–1 scale. The parameters with the strong correlations are denoted by 1, whereas those with weak correlations between the absolute GRG and the SSGRG are denoted by 0. Deng GRG has a standardized with a range of 0.5–1. The absolute and second synthetic GRGs are both in the range of 0 to 1. The strongest correlations found between each of the four North African countries that were matched with an SDG are listed below on the basis of the absolute GRG values.

Morocco has demonstrated the highest weights of 0.925369, 0.96191, 0.970693, 0.9584, 0.8315 and 0.925369 for SDG1, SDG2, SDG7, SDG9, SDG15 and SDG16, respectively, as presented in Table 2 and Fig. 7. SDG1 is predicted to generate a considerable level of influence since the pandemic has resulted in the downsizing of employees and the shutdown of businesses, which has decreased household income (Martín–Blanco et al., 2022). Nevertheless, according to the national review of the implementation of the SDGs, the societal effects are substantial (2021). While the associative actors who participated in the consultation acted in favor of all 17 SDGs, they placed the most emphasis on SDG4 in the promotion of education (44.6 %), followed by SDG1 in the fight against poverty (33.1 %), and SDG5 in the achievement of gender equality (32.2 %).

Table 2
WDI and Social Sustainable Goals.

SDGs #	Countries	Absolute GRG	Deng GRG	SSGRG
SDG 1	Algeria	0.921269	0.601740	0.761504
	Tunisia	0.770920	0.587973	0.679446
	Egypt	0.913610	0.560770	0.737191
	Morocco	0.925369	0.658320	0.791844
	Sudan	0.672412	0.517520	0.594966
SDG 2	Algeria	0.747020	0.584400	0.665719
	Tunisia	0.899260	0.558470	0.728865
	Egypt	0.960940	0.613910	0.787422
	Morocco	0.961910	0.644000	0.802955
	Sudan	0.692180	0.612310	0.652245
SDG 3	Algeria	0.997300	0.747422	0.872361
	Tunisia	0.989483	0.731870	0.860676
	Egypt	0.980380	0.713700	0.847040
	Morocco	0.963010	0.711180	0.837095
	Sudan	0.754280	0.669012	0.711646
SDG 4	Algeria	0.937090	0.849640	0.893365
	Tunisia	0.869190	0.786340	0.827765
	Egypt	0.775890	0.752540	0.764215
	Morocco	0.814698	0.784540	0.799619
	Sudan	0.649012	0.798510	0.723761
SDG5	Algeria	0.984279	0.615623	0.799951
	Tunisia	0.962728	0.666134	0.814431
	Egypt	0.850469	0.453577	0.652023
	Morocco	0.675783	0.534528	0.605155
	Sudan	0.610923	0.520971	0.565947
SDG 10	Algeria	0.915360	0.864810	0.890085
	Tunisia	0.941469	0.613063	0.777266
	Egypt	0.962689	0.618238	0.790463
	Morocco	0.937369	0.608963	0.773166
	Sudan	0.770127	0.526543	0.648335
SDG16	Algeria	0.921269	0.950079	0.935674
	Tunisia	0.770920	0.962728	0.866824
	Egypt	0.913610	0.850469	0.882039
	Morocco	0.925369	0.675783	0.800576
	Sudan	0.730012	0.520914	0.625463
SDG17	Algeria	0.900300	0.737600	0.818950
	Tunisia	0.889600	0.744400	0.817000
	Egypt	0.631300	0.677100	0.654200
	Morocco	0.885500	0.740300	0.812900
	Sudan	0.610500	0.609870	0.610180

Algeria has correspondingly shown the same level of strong association for SDG3, SDG4, SDG5, SG11, SDG13, SDG14 and SDG17, with weights of (0.9973), (0.93709), (0.984279), (0.88589), (0.899977), (0.958205) and (0.9003), respectively. Algeria shows superior performance in terms of environmentally sustainable goals for SDG 13 and SDG14, with values of 0.7637505 and 0.7671385, respectively, as shown in Table 3 and Fig. 8.

However, Egypt has only one strong association (0.6228), which is for SDG8. The weakest associations in the absolute GRG were (0.96301), (0.675783), (0.5832), (0.773166), (0.855978), and (0.71564) for SDG3, SDG5, SDG8, SDG11, SDG12 and SDG14, respectively, in Morocco. Investing an anticipated 30 billion dirhams, the established electrical power from renewable sources will expand to 12,896 MW in 2030 [57]. In 2015, a new initiative was initiated with the purpose of levelling the socioeconomic field across various classes of society and between areas in terms of infrastructure and fundamental social facilities in rural regions to eventually consolidate efforts toward the implementation of SDG9 (Ibn Batouta et al., 2024). Moreover, the Digital Morocco 2020 strategy seeks to encourage innovation through democratizing digital usage in the private industry as well as among community members by decreasing the web access divide by 50 %, connecting 20 % of SMEs to the online industry, and investing in quality mobile communication

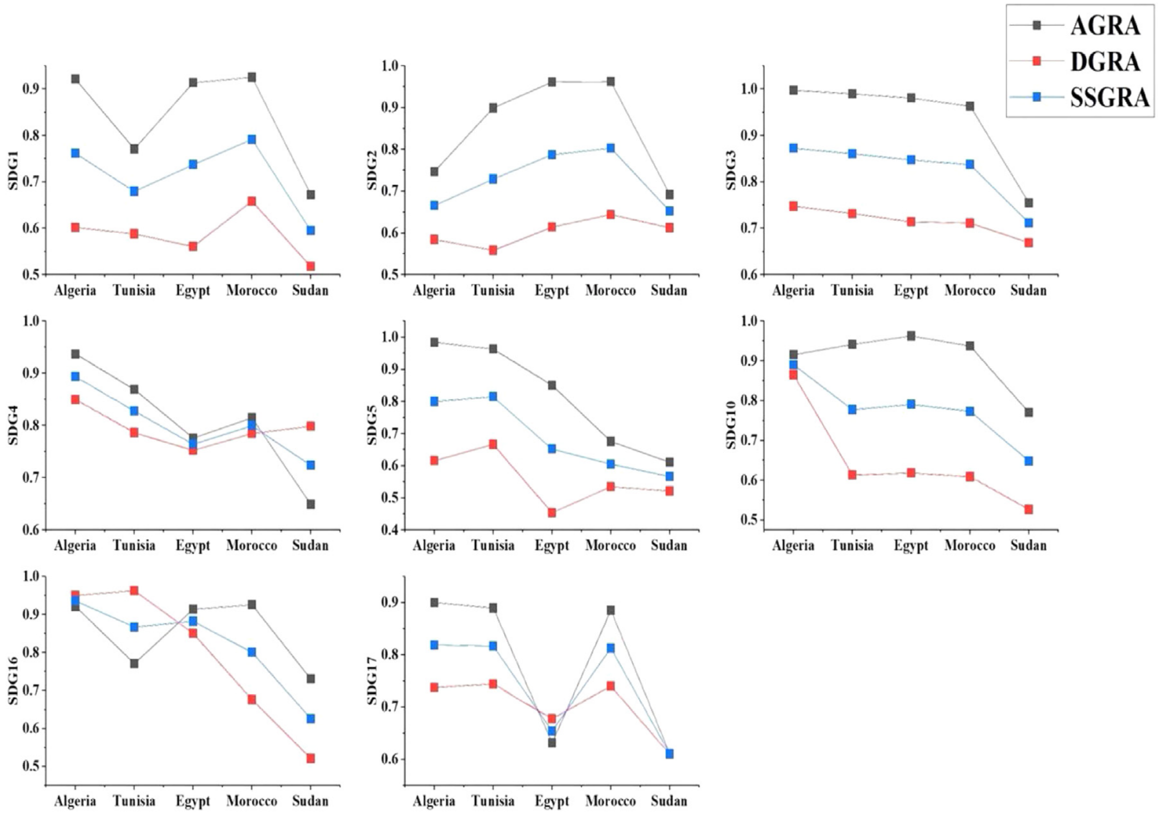

Fig. 7. Nexus of WGI and social sustainability in North Africa.

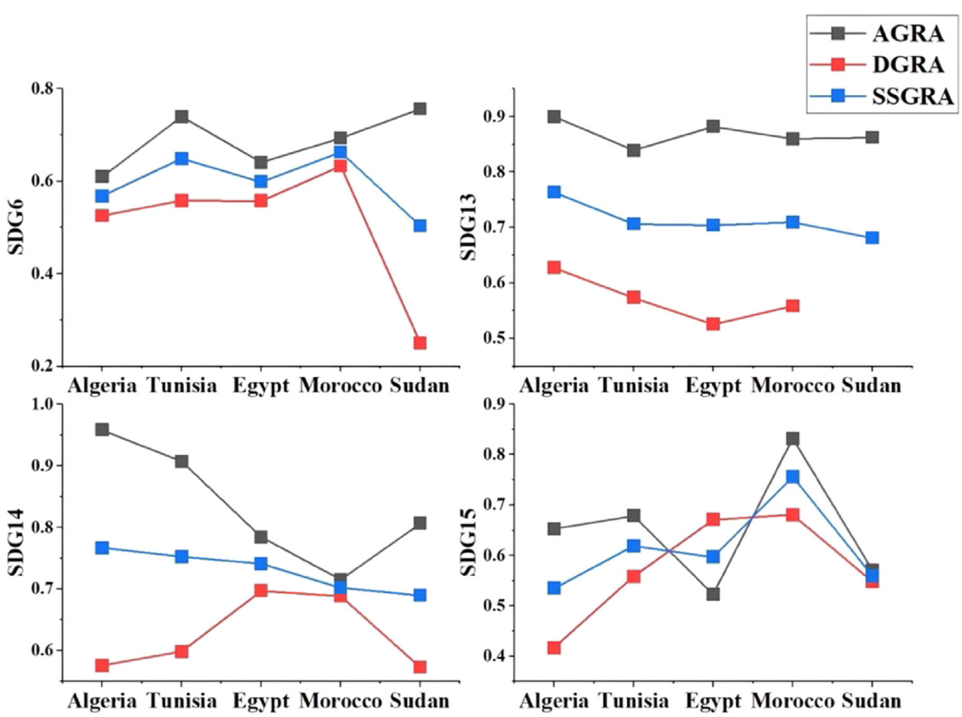

Fig. 8. Nexus of WGI and environmental sustainability in North Africa.

infrastructures across the country (Industrial Acceleration Plan, 2020). Nevertheless, Morocco's research and development budget is severely constrained.

There have been successes, but there are also obstacles that must be overcome to realize the SDGs in the health sector. One such example is the creation of a health-related data system to track the leading causes of mortality and report on the relevant SDG indices. In addition, accelerating the decline in avoidable mortality and morbidity associated with social variables such as poverty is critical. Despite Morocco's considerable improvements, the country still faces several obstacles in regard to protecting maritime environments, which eventually influences advancements toward SDG13 [58]. For effec-

Table 3
WDI and Environmental Sustainable Goals.

SDGs #	Countries	Absolute GRA	Deng GRA	SSGRA
SDG 6	Algeria	0.609842	0.525736	0.567765
	Tunisia	0.739724	0.558396	0.649058
	Egypt	0.640603	0.557626	0.599113
	Morocco	0.693357	0.633039	0.663194
	Sudan	0.756642	0.250003	0.503321
SDG 13	Algeria	0.899977	0.627524	0.763750
	Tunisia	0.839128	0.573737	0.706432
	Egypt	0.882253	0.525731	0.703991
	Morocco	0.859672	0.558396	0.709034
	Sudan	0.862468	0.500004	0.681234
SDG 14	Algeria	0.958205	0.576072	0.767138
	Tunisia	0.906953	0.598382	0.752667
	Egypt	0.784657	0.697412	0.741034
	Morocco	0.715640	0.688883	0.702261
	Sudan	0.806824	0.573200	0.690012
SDG 15	Algeria	0.653113	0.417167	0.535135
	Tunisia	0.678826	0.558360	0.618592
	Egypt	0.522904	0.671205	0.597054
	Morocco	0.831507	0.680308	0.755910
	Sudan	0.571205	0.548124	0.559660

tive national coast management, national authorities must be reinforced to address challenges such as pollution from all origins to which the coast is vulnerable and illegal fishing. In Algeria, SDG2 (0.747020), SDG6 (0.609842), and SDG10 (0.958605) represent the least common associations. For the remaining SDG4, SDG9, SDG15, and SDG17 indicators, the weakest relationships are detected in Egypt, with weights of (0.775890), (0.612503), (0.522904) and (0.631300), respectively. Tunisia's lowest values were for SDG1 (0.770920), SDG7 (0.872385), SDG13 (0.839128), and SDG16 (0.770910). For Morocco, the Deng GRG had the strongest influence on SDG1 (0.658320), SDG2 (0.644000), SDG6 (0.633039), SDG7 (0.697412), SDG9 (0.960507), SDG13 (0.874847), and SDG15 (0.680308) and the least influence on SDG 3 (0.711180), SDG 10 (0.608963), and SDG16 (0.675783). For Algeria, the greatest effect was associated with SDG3 (0.747422), SDG4 (0.849640), and SDG11 (0.940026), whereas the weak influence was associated with SDG6 (0.525736), SDG8 (0.633039), SDG10 (0.864810), SDG14 (0.576072), and SDG15 (0.417167). The degree of impact was strong in Tunisia with respect to SDG5 (0.666134), SDG16 (0.962728), and SDG17 (0.744400). This impact was the lowest for SDG2 (0.558470) and SDG7 (0.580272). In contrast, Morocco showed exceptional economic sustainability performance in the regions of SDG7, SDG8, SDG9 and SDG12, with SSGRGs of 0.834052, 0.8340525, 0.959450 and 0.865409, respectively, as presented in Table 4 and Fig. 9.

Furthermore, Sudan demonstrates the least effective performance in relation to the SDGs among all North African nations.

Sudan's achievement of the SDGs has been significantly hindered by economic difficulties. The country faces rising inflation, external debt, and limited access to international aid due to sanctions and insufficient foreign investment [59]. Economic volatility has adversely impacted SDG 1 (No Poverty), SDG 2 (Zero Hunger), and SDG 10 (Reduced Inequality) by intensifying poverty, inequality, and food insecurity. United Nations Economic Commission for Africa, [25] indicated that Sudan's insufficient funding for SDG-related activities is a consequence of its restricted budgetary capability, exacerbating the problem. The sluggish advancement in attaining SDG objectives has been attributed to the deficiency of social services and infrastructure in rural areas. Finally, Egypt SDGs 1, 4, 5, 8, 9, 11, 13 and 17 had the weakest effects, with weights of 0.560770, 0.560760, 0.752540, 0.453577, 0.631300, 0.614609, 0.839128, 0.525731, and 0.677100, respectively. The rankings of all North African countries on the basis of the SSGRA are presented in Table 5.

The ranking of each grey model has been used to evaluate how countries perform concerning the relevant SDGs. On the basis of the results,

Table 4
WDI and Economic Sustainable Goals.

SDGs #	Countries	Absolute GRG	Deng GRG	SSGRG
SDG 7	Algeria	0.964195	0.633039	0.798617
	Tunisia	0.872385	0.580272	0.726328
	Egypt	0.888579	0.598382	0.743480
	Morocco	0.970693	0.697412	0.834052
	Sudan	0.779812	0.508712	0.644262
SDG 8	Algeria	0.964195	0.633039	0.798617
	Tunisia	0.872385	0.580272	0.726328
	Egypt	0.888579	0.598382	0.743480
	Morocco	0.970693	0.697412	0.834052
	Sudan	0.790434	0.519729	0.655081
SDG 9	Algeria	0.820304	0.922407	0.921350
	Tunisia	0.869107	0.871203	0.870150
	Egypt	0.612503	0.614609	0.613550
	Morocco	0.958406	0.960507	0.959450
	Sudan	0.528722	0.540918	0.534815
SDG 11	Algeria	0.885894	0.940026	0.912955
	Tunisia	0.777266	0.871256	0.824258
	Egypt	0.790463	0.839128	0.814795
	Morocco	0.773166	0.867153	0.820158
	Sudan	0.701712	0.670921	0.686316
SDG 12	Algeria	0.936866	0.732548	0.834707
	Tunisia	0.984369	0.711099	0.847734
	Egypt	0.901680	0.803392	0.852536
	Morocco	0.855978	0.874847	0.865409
	Sudan	0.750920	0.679612	0.715266

Table 5
Grey assessment rankings of North African countries.

Variables	Grey Rational Model	Ranking
WGI and SDG 1	**Second Synthetic Grey Relational Analysis Model (SSGRA)**	Morocco>Algeria>Egypt>Tunisia>Sudan
WGI and SDG 2		Morocco>Egypt>Tunisia>Algeria>Sudan
WGI and SDG 3		Algeria>Tunisia>Egypt>Morocco>Sudan
WGI and SDG 4		Algeria>Tunisia>Morocco>Egypt>Sudan
WGI and SDG 5		Tunisia>Algeria>Egypt>Morocco>Sudan
WGI and SDG 6		Morocco>Tunisia>Egypt>Algeria>Sudan
WGI and SDG 7		Morocco>Algeria>Egypt>Tunisia>Sudan
WGI and SDG 8		Algeria>Tunisia>Morocco>Egypt>Sudan
WGI and SDG 9		Morocco>Algeria>Tunisia>Egypt>Sudan
WGI and SDG 10		Algeria>Tunisia>Morocco>Egypt>Sudan
WGI and SDG 11		Algeria>Tunisia>Morocco>Egypt>Sudan
WGI and SDG 12		Morocco>Egypt>Tunisia>Algeria>Sudan
WGI and SDG 13		Algeria>Morocco>Tunisia >Egypt>Sudan
WGI and SDG 14		Algeria>Tunisia >Egypt>Morocco>Sudan
WGI and SDG 15		Morocco>Tunisia>Egypt>Algeria>Sudan
WGI and SDG 16		Algeria>Egypt>Tunisia >Morocco>Sudan
WGI and SDG 17		Tunisia >Algeria>Morocco>Egypt>Sudan

Morocco has maintained its 1st ranking with SDG1, SDG2, SDG7, SDG9, SDG12 and SDG15, the same as with absolute GRG and SSGRG, which confirms the strong association of the combined effect of COVID-19 and the WGI index with the goals of interest. For SDG6, Morocco was the first in the ranking and the SSGRG. Morocco is acutely aware of the severity of its water shortage, and many initiatives have been launched to address this problem [60]. Despite progress, the high price tag of unconventional water mobilization programs remains a barrier that must be overcome via the concerted efforts of all relevant parties and increased financial support [61]. However, the results of the absolute GRG analysis revealed that Tanisia had the highest weight. The absolute GRG, Deng GRG and SSGRG results were the same as those of Algeria, with SDG3, SDG4, SDG11, and SDG13 being the first-ranked indicators. The country also had the first ranking in SDG8, SDG10, SDG14 and SDG16 as well as the highest values in the SSGRG; however, in the absolute GRG, Egypt had the strongest associations with SDG8 and SDG10, and for SDG 16, Morocco had the highest weight. In contrast, Tunisia scored first with SDG5 and SDG17, followed by Algeria, while having the strongest association according to the absolute GRG. Egypt had the last ranking

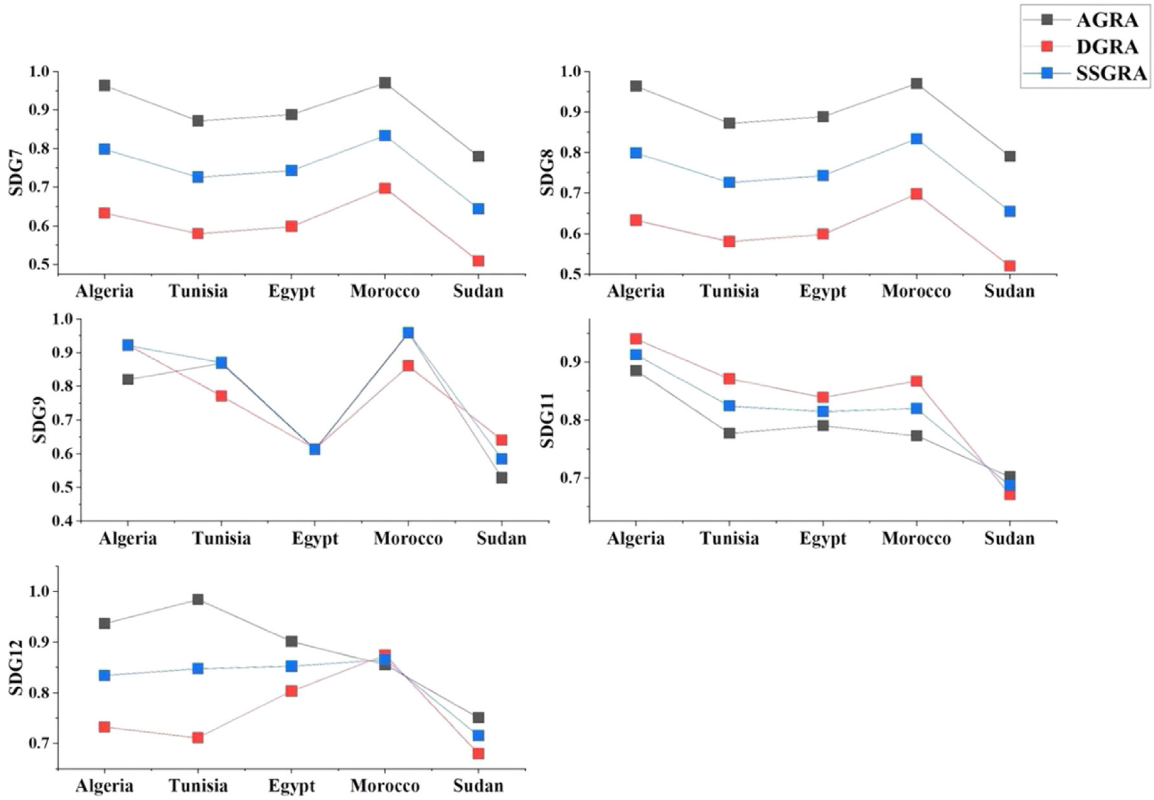

Fig. 9. Nexus of WGI and economic sustainability in North Africa.

in SDG4, SDG8, SDG9, SDG11, SDG13 and SDG17, which completely matches the weakest associations in the SSGRG results. However, the Sudan scores for all the SDGs were low among all the countries in North Africa. Except for SDG10, where Egypt had the highest values of absolute GRG results.

Moreover, this decision analysis assisted in measuring the differences in the WGI and SDGs among North African countries. With this aim, an initial step is essential to assess the criteria and alternative actions categorized in Table 6. where p = 4, n = 17, and result = v (a_i, S_j) while i = 1, 2, 3,4 and j = 1, 2, 3, 4, 5….17. Let s1, s2, s3…s17 characterize the strongest grey relation analysis of WGI on all SDGs in the four North African countries. Table 6 presents the SSDGRA degree of the grey relation-based matrix, which illustrates the second synthetic grey relations between the decision criteria and the decision actions. To quantify the effect of WGI on SDG achievement in North African nations, we established a decision-making framework for comparison analysis.

Table 6
Decision Parameters.

Goal	Measuring Grey Relation (Association) Between WGI and SDGs 1–17 within North African Countries
State of Nature/Criteria (Sj); N = 1, 2,…,k	SDG01 (S01) SDG02 (S02) SDG03 (S03) SDG04 (S04) SDG05 (S05) SDG06 (S06) SDG07 (S07) SDG08 (S08) SDG09 (S09) SDG10 (S10) SDG11 (S11) SDG12 (S12) SDG13 (S13) SDG16 (S14) SDG15 (S15) SDG16 (S16) SDG17 (S17)
Alternatives (a_i); i = 1, 2,…m	Grey Relation was superior between WGI and SDGs performance in Algeria ($\mathbf{a_1}$) Grey Relation was superior between WGI and SDGs performance in Tunisia ($\mathbf{a_2}$) Grey Relation was superior between WGI and SDGs performance in Egypt ($\mathbf{a_3}$) Grey Relation was superior between WGI and SDGs performance in Morocco ($\mathbf{a_4}$) Grey Relation was superior between >WGI and SDGs performance in Sudan ($\mathbf{a_5}$)

Table 7
Criteria action matrix.

SSGRA	a_1	a_2	a_3	a_4	a_5
S1	0.761504	0.679446	0.737190	0.791844	0.661241
S2	0.665710	0.728865	0.787420	0.802955	0.501231
S3	0.872361	0.860676	0.847040	0.837095	0.542310
S4	0.893365	0.827765	0.764215	0.799619	0.661241
S5	0.799951	0.814431	0.652023	0.605155	0.560914
S6	0.567765	0.649058	0.599113	0.663194	0.503321
S7	0.798617	0.726328	0.743480	0.834052	0.629815
S8	0.770450	0.738450	0.627050	0.734350	0.660719
S9	0.921350	0.870150	0.613550	0.959450	0.515670
S10	0.890085	0.777266	0.790463	0.773166	0.706812
S11	0.912955	0.824258	0.814795	0.820158	0.689456
S12	0.834707	0.847734	0.852536	0.865409	0.729385
S13	0.763750	0.706432	0.703991	0.709034	0.681234
S14	0.767138	0.752667	0.741034	0.702260	0.690012
S15	0.535135	0.618590	0.597050	0.755900	0.504321
S16	0.935674	0.866824	0.882039	0.800576	0.750932
S17	0.818950	0.817000	0.654200	0.812900	0.592980

Moreover, a decision-making process model grounded in SSGRA is applied for the comparative analysis to examine the effect of WGI on SDG performance in North Africa, as depicted in Table 7.

4.1. Conservative (maximin) model analysis

Fig. 10 depicts the GRA-observed superior SDG performance in WDI in Algeria, Tunisia, Egypt, Morocco and Sudan. Among the 17 SDGs, SDG9, SDG16, SG11, SDG12, SDG10, and SDG4 had the highest performance in North Africa. Moreover, we employed a method that has been effectively applied in several investigations (Ikram et al., 2019). This is known as a maximin criterion since we optimize the SDG performance function.

$$max\ Oi\ \{min\ O_k V\left(O_i, X_j\right)\} = min\ O_p \begin{bmatrix} 0.791844 \\ 0.802955 \\ 0.872361 \\ 0.893365 \\ 0.814431 \\ 0.663194 \\ 0.834052 \\ 0.770450 \\ 0.959450 \\ 0.890085 \\ 0.912955 \\ 0.834707 \\ 0.763750 \\ 0.767138 \\ 0.755900 \\ 0.935673 \\ 0.818950 \end{bmatrix} = 0.6631945\ (Morocco, SDG6)$$

The results reveal that the most influential country's SDG performance by ineffective WGI among all North African nations is Morocco. The function outcome also shows that SDG6 (access to clean water and sanitation) is the most impacted goal of all 17 SDGs in Morocco. This can be explained by the current state of water emergence in Morocco.

Morocco has been facing reduced water availability due to severe drought. In March 2022, the country endured its most severe drought over four decades [62]. In the midst of worldwide heatwaves and little precipitation, North Africa struggles to provide viable water access in water-scarce areas, undermining the nation's primary economic sector, agriculture [63]. Furthermore, intensive agricultural methods, pollution, and rapid urbanization have significantly strained surface water resources. This potentially results in adverse effects on the physical and chemical properties of water.

Morocco has transitioned from a state of water scarcity to a state of severe water strain, with projections suggesting a 30 % reduction in its water assets by 2050. Climate change has had a direct influence on Morocco's agriculture, which heavily relies on rainfall and its food supply [64]. The World Resources Institute (WRI) rated Morocco 22nd among the nations that are most in danger from water scarcity [65]. Morocco enacted a National Water Plan (PNE) in 2020, outlining an extensive action plan for investing up to $40 billion designated for the water industry to restore and maintain water availability [61].

Moreover, according to a 2015 UN report, there was a single, obvious truth that underlined every obstacle to resolving the water issue. This is also due to the concerns of national entities, who are not aware of the need for water conservation and efficient water usage to increase water efficiency in their regions. Largely poor individuals, specifically poor women, are those most affected by water and sanitation crises, and they frequently lack the political voice necessary to assert their rights to water (Vogel et al., 2022). Thus, Voice & Accountability contributes to the slowdown of SDG performance in North Africa, which explains the strong association demonstrated in our findings. Our results help provide North African national authorities with a clear vision of which SDGs to prioritize. Evidently, all water, and sanitation-related SDGs exist. In addition, the conclusions of the study call for action to address North African citizens' freedom of expression and freedom of choosing their government through the implementation of pertinent policies. This eventually contributed to the enhancement of the WGI index in North Africa as well as the acceleration of the SDGS performance.

5. Discussion

The GRA results highlight a complex relationship between governance quality and sustainable development outcomes in North Africa. Overall, the region shows comparatively strong performance on several SDGs notably SDG 4 (quality education), SDG 9 (industry, innovation, and infrastructure), SDG 10 (reduced inequalities), SDG 11 (sustainable cities and communities), SDG 12 (responsible consumption and produc-

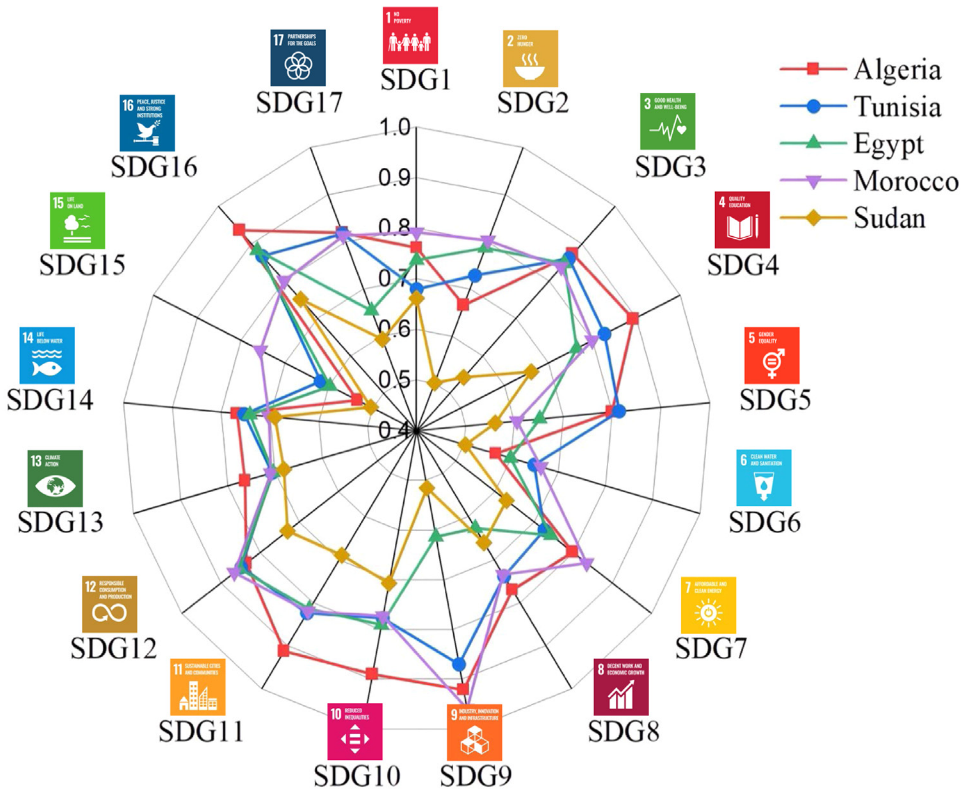

Fig. 10. Grey relationship superiority between WGI and SDG performance in North Africa.

tion), and SDG 16 (peace, justice, and strong institutions). These SDGs had the highest composite scores in the SSGRA, suggesting that North African countries have made notable progress in education access, infrastructure development, urban sustainability, and institutional frameworks. This pattern implies that wherever governance has effectively channeled resources and policies for example, into schooling or city planning tangible development gains have followed. By contrast, other goals remain challenging. The analysis identifies SDG 6 (clean water and sanitation) as the most pressing laggard across the region, aligning with the well-documented water scarcity crisis in North Africa.

Despite some improvements, basic needs like water, along with persistent issues in poverty (SDG 1), gender equality (SDG 5), decent work (SDG 8), and inequality (SDG 10 in certain contexts), continue to be difficult to fully address. This juxtaposition of strengths and weaknesses reveals that sustainable development in North Africa is highly sensitive to the quality of governance in each sector. Progress has not been uniform it mirrors governance successes in some areas and governance shortfalls in others, underlining the critical role institutions play in driving or hindering each SDG. Zooming in to country-specific results, a complex picture of governance dynamics emerges.

Morocco stands out in the analysis: its governance indicators show the strongest associations with SDG 1 (no poverty), SDG 2 (zero hunger), SDG 7 (affordable and clean energy), SDG 9 (industry and infrastructure), SDG 15 (life on land), and SDG 16 (institutions). This suggests that improvements in governance from more effective public services to better rule enforcement have substantially benefited Morocco's outcomes in poverty alleviation and infrastructure, as well as in energy and environmental management. Indeed, Morocco's proactive policies in renewable energy (under its national energy strategy since 2009) likely contributed to the exceptionally high SDG 7 performance . However, for SDG7, starting in 2009, following the High Royal Instructions, Morocco implemented an energy plan to meet its requirements and expand its renewable resources [66]. Strategically, it aims to increase regional cooperation and promote sustainable development through the promotion of alternative energy sources, improvements in energy efficiency, and advancements in reliable and competitive technologies [67].

Likewise, strong institutional commitment to industrial development and anti-poverty programs has paid dividends in SDG 9 and SDG 1. However, Morocco's achievements are counterbalanced by critical challenges, particularly regarding SDG 6. The GRA identified clean water and sanitation as Morocco's most problematic goal, with water access and quality showing a weak governance linkage relative to other SDGs.

In practice, this means that despite Morocco's overall above-average SDG index score in the region, water scarcity and sanitation shortfalls are undermining its sustainable development progress. Recurrent droughts and infrastructure gaps have led to frequent water shortages, public complaints about supply cuts, and threats to the agricultural sector, which depends on increasingly unreliable water resources. These findings reveal a governance gap in managing natural resources: while Morocco has established strategies and even managed to stay on schedule in policy intentions for water access), implementation lags due to climate pressures and perhaps insufficient institutional capacity or priority given to water governance. In summary, Morocco's case illustrates how targeted governance efforts can yield strong gains in certain SDGs, but also how a single crucial sector like water can become a binding constraint on sustainable development if not governed effectively. Algeria's governance–SDG profile shows a different mix of strengths and weaknesses. The study finds Algeria had very strong governance links to SDG 3 (good health and well-being), SDG 4 (quality education), SDG 5 (gender equality), SDG 11 (sustainable cities), SDG 13 (climate action), SDG 14 (life below water), and SDG 17 (partnerships). Projects such as the Moroccan Solar Plan (MSP) and the Combined Wind Program (IWP) for integrated solar and wind energy generation are now available. In 2020, Morocco is planning to have renewable energy accounting for 42 % of total electricity consumption, and by 2030, concerned authorities wish to have reached 52 % (Aida Alami, 2021).

Algeria's state institutions have achieved solid outcomes in public health, education, and certain environmental protections. For instance, despite pandemic-related setbacks, Algeria managed to maintain educational continuity to some extent and shows commitment to climate and marine sustainability, as reflected in relatively high SDG 13 and SDG 14 scores. These successes likely stem from Algeria's centralized governance model, which, while authoritarian in nature, has historically invested in social programs and infrastructure using its hydrocarbon revenues. The GRA suggests that where Algerian governance is effective and stable such as delivering basic education or coordinating climate action – the corresponding SDG outcomes are positive. On the other hand, Algeria exhibits weaker governance associations with SDG 2 (zero hunger), SDG 6 (clean water), and SDG 10 (reduced inequalities). With respect to SDG 4, COVID-19 pushed the switch to an online educational system, but owing to limited digital educational infrastructure, it was not adequately prepared for this mandate (Crawford & Cifuentes-Faura, 2022).

Notably, a relatively low correlation with SDG 6 implies Algeria also struggles with water management, possibly in rural areas lacking adequate sanitation or in balancing agricultural water use, though not as acutely as Morocco. The weak linkage to SDG 2 and SDG 10 points to persistent inequality and inclusion issues despite Algeria's national wealth, pockets of poverty and regional disparities remain. Governance factors such as bureaucratic inefficiencies or corruption in welfare programs could be impeding the eradication of hunger and the reduction of inequality. In essence, Algeria's governance dynamics reveal a state capable of grand projects and maintaining macro-level services (hospitals, schools, infrastructure) but less agile in ensuring equitable outcomes and resilience in basics like food security and local water provision. This dichotomy resonates with broader observations that economic resources alone are not enough without accountable and inclusive governance to distribute those resources effectively. Egypt's results are perhaps the most stark in underlining the importance of governance. The analysis found only one strong governance–SDG association for Egypt: SDG 8 (decent work and economic growth).

Egypt's recent governance efforts have chiefly translated into economic growth-related outcomes, while progress on other goals shows a much weaker governance imprint. Over the past decade, Egypt's government has prioritized large-scale economic initiatives – from infrastructure megaprojects to business reforms which likely explains the notable alignment with SDG 8. Indeed, even amid global turbulence, Egypt maintained moderate GDP growth and undertook measures to stabilize the economy. The slowing expansion of the economy is to blame for this, with GDP growth of 3.6 % in FY2019/2020 [68]. Moreover, for the first time since its creation in 1990, the human development index declined significantly in 2015 [69].

However, the limited scope of strong SDG linkages points to shortcomings in how governance improvements are extending to social and environmental domains. Egypt's weaker governance influence on SDGs such as quality education, industry and innovation, life on land, and partnerships (SDGs 4, 9, 15, 17) suggests a development model that has not been inclusive or holistic. This is consistent with Egypt's human development trends – for instance, a notable drop in Egypt's Human Development Index was recorded around 2015, reflecting stagnation or decline in health and education outcomes despite economic growth. Entrenched issues like high corruption, constrained civil liberties, and centralization of power may be undermining broader SDG progress. Without robust accountability and institutional transparency, economic gains have not fully translated into poverty reduction, quality education, or environmental sustainability. Egypt's case reveals a critical governance dynamic: a state can exert control to drive GDP growth and infrastructure, but unless governance reforms also empower communities, curb corruption, and strengthen public institutions, improvements in livelihoods and environmental quality will remain limited. This aligns with the idea that sustainable development requires governance that is not only strong in capacity but also inclusive and accountable.

The Egyptian government's pandemic responses such as financial aid to workers and support for industries show that targeted governance interventions can mitigate shocks; yet the enduring challenge is to broaden such efforts to systematically advance education, equity, and ecological conservation. Nonetheless, after COVID-19, the Egyptian economy is predicted to soon rebound due to domestic growth, which leaves the nation less vulnerable to worldwide economic downturns. In response to the pandemic, Egypt implemented proactive measures to mitigate the adverse effects on various sectors. They distributed financial assistance and conducted training programs for over 6 million both formal and informal workers [68]. Furthermore, they eased financial constraints for the shipping, aviation, and tourism industries, as well as for SMEs [68].

Tunisia offers a contrasting scenario as a country that underwent significant political change. The GRA findings indicate that Tunisia's governance had especially strong effects on SDG 5 (gender equality), SDG 16 (peace, justice, and strong institutions), and SDG 17 (partnerships for the goals). These are goals closely tied to institutional openness and civic participation, reflecting how Tunisia's post-2011 democratization initially fostered improvements in women's rights, governance transparency, and international cooperation. For example, Tunisia implemented progressive gender equality laws and a new constitution that strengthened civil liberties, likely contributing to its above-regional performance on SDG 5. The high association with SDG 16 underscores that voice and accountability, judicial independence, and anti-corruption measures in Tunisia translated into better governance outcomes in fact, Tunisia had been regarded as a regional leader in governance indicators like voice and accountability during the 2010s. However, Tunisia's weaker governance linkages to SDG 2 (hunger) and SDG 7 (energy) reveal that democratization alone did not immediately solve economic vulnerabilities. The country has faced persistently high unemployment and rural poverty, which governance improvements have yet to fully tackle. Likewise, energy (SDG 7) remains an issue, as Tunisia still relies heavily on imports and has been slow to diversify energy sources compared to Morocco. These shortfalls suggest that institutional quality needs to be matched with effective economic management and resource governance. Tunisia's recent political turbulence and economic woes (especially post-2020) serve as a reminder that early gains in governance can be fragile. Nevertheless, the strong performance on institutional SDGs in Tunisia's case affirms the broader finding that good governance – particularly in the form of democratic accountability and strong institutions can drive social progress (like gender equality and rule of law) even if economic dividends take longer to materialize. It highlights a critical dynamic: North African countries that have invested in governance reforms (even amid political upheavals) tend to see positive social outcomes, whereas neglecting such reforms can stall progress on those fronts.

Finally, the inclusion of Sudan (though often categorized in sub-Saharan Africa, it shares many North African contextual challenges) in the analysis underscores how severe governance deficits correspond to poor SDG performance. Sudan consistently ranked at or near the bottom on all six WGI dimensions in the region, and its sustainable development outcomes mirror this reality. The country's protracted conflicts, political instability, and weak institutions have left it struggling on nearly every SDG, from poverty and hunger to health and infrastructure. Political instability, economic challenges, and ongoing conflicts were significant factors hindering Sudan's progress in the SDGs from 2016 to the present. Significant political upheaval has occurred in the nation, which faces challenges in enacting meaningful governance reforms [70]. Kuol, [71] contend that ongoing internal conflicts have redirected resources away from social development projects essential for realizing the SDGs. Political instability has not only weakened institutions but also hindered the government's capacity to address essential issues such as healthcare, education, and poverty alleviation. Unstable governance and uncertain environments persistently hinder progress considerably.

For instance, Sudan's overall SDG Index ranking is among the lowest globally , reflecting chronic shortfalls in basic services and rights. The study's results likely show very low grey relational grades for Sudan across most goals, signifying that without minimum governance functionality (security, rule of law, administrative capacity), sustainable development cannot take hold. This extreme case reinforces the central message of the findings: governance quality and sustainable development are deeply intertwined, to the point that critical governance failures (as in Sudan) can negate progress on nearly all development fronts. Conversely, countries with even moderate improvements in governance (as seen in the varying degrees for Morocco, Algeria, Tunisia, and Egypt) exhibit correspondingly better outcomes. In a broader literature context, this aligns with robust evidence from Africa and beyond that good governance is a prerequisite for sustainable development.

Prioior studies have shown that across African countries, governance quality measured through indices similar to the WGI has a strong, positive effect on economic, social, and environmental sustainability. The North African experience detailed in this study adds rich nuance to that narrative: it shows which aspects of development are most sensitive to governance in specific national contexts. In summary, the findings reveal a mosaic of governance dynamics in North Africa. They demonstrate that effective, accountable governance can catalyze progress in education, infrastructure, equality, and environmental stewardship, as evidenced by the high SDG achievements where WGI scores are strong. At the same time, persistent governance gaps whether in managing water resources, ensuring inclusivity, or curbing corruption can significantly impede SDG attainment, creating critical vulnerabilities like the water crisis affecting Morocco's SDG 6. The associations between specific WGI dimensions and SDGs suggest that each facet of good governance plays a role: for example, rule of law and regulatory quality appear to underpin economic and infrastructure gains, while voice and accountability relates to social inclusion and equity, and control of corruption spans all domains to ensure resources reach their intended targets.

North Africa's trajectory toward the 2030 Agenda thus clearly hinges on governance: the more its countries strengthen their institutions, engage their citizens, and uphold the rule of law, the more likely they are to accelerate sustainable development. This study's granular insights into the governance SDG nexus serve as an empirical confirmation that investing in governance reforms is not a separate agenda from achieving the SDGs, but rather a fundamental enabler of success across all development goals

6. Conclusion and policy implications

This study aims to address this gap in the literature by analyzing the impact of WGI on SDG performance in North African countries. The data analysis covers the years 2016 and 2023. For our analysis, we employed a novel integrated grey relational model to assess and analyze the impact of WGI on the SDGs of North African countries. Our findings demonstrate that the performance of the SDGs is mostly affected in Morocco. The presented weights revealed strong evidence between WGI and SDG1, SDG2, SDG7, SDG9, SDG15 and SDG16 in Morocco. Furthermore, SDG6, i.e., access to and availability of water and sanitation, was the most impacted SDG among all 17 goals among the four North African countries. The results align with the current situation in Morocco, since water has become scarce and Moroccan citizens have started complaining about continuous water cuts in their regions. The contributing factor to this alarming case is the large changes in climate change, which limited rainfall around the kingdom and threatened the primary sector of Morocco, which is agriculture. Our study is a call for action to the Moroccan leader to take into account sensitive SDGs, which urges immediate intervention to slow their influence and accelerate their performance through planning a relevant roadmap for the next year, 2023. Our research provides North African businesses and organizations with a framework for analyzing the long-term effects of their operations, prompting them to consider how they may make a constructive contribution to society. As a result, they can comprehend their role in achieving the desired outcome, track their progress toward the objective, and share information about their progress. This will provide more information, reporting and insights to North Africa policymakers in developing a more holistic perspective from which to implement the SDGs successfully. Furthermore, it is suggested that countries in which the WGI is very high be studied to analyze the way their SDGs are positively impacted and from which governments with low voices and accountability indices can derive practical recommendations and show more seriousness regarding this index.

Moreover, an important avenue for future work is to integrate digital technology into the governance-sustainability nexus. Digital tools and advanced analytics can greatly enhance data-driven decision-making for sustainable development. For instance, recent studies have employed sophisticated fuzzy logic and multi-agent systems to evaluate clean energy systems, showcasing how technology can manage complexity and uncertainty in sustainability contexts. Similarly, applying big data analytics or AI to governance and SDG datasets could uncover nonlinear patterns and provide real-time monitoring of progress. We suggest that future research explore such digital governance platforms for example, how e-government systems, open data, and machine learning models might improve the implementation of SDG-related policies. Emphasizing digital technology in this field could lead to more robust, adaptive, and transparent frameworks for achieving sustainable development goals.

Additionally, future research may include other African countries to investigate the relationships between WGI and the SDGs. The empirical results reported here should be considered in light of several limitations. The primary limitation to the generalizability of these results is the lack of previous studies investigating the effects of WGI on SDG performance. Future research could consider other indicators that may affect the performance of the SDGs in the African region.

6.1. Managerial implications

The findings carry important implications for policymakers and institutional leaders in North Africa. Governance improvements should be viewed as a strategic priority on par with direct investments in health, education, or infrastructure, because the analysis shows that stronger governance drives better outcomes across those sectors. For national governments, this means that initiatives to strengthen rule of law, enhance transparency, and control corruption are not abstract political reforms they are practical steps toward accelerating SDG progress. Leaders need to internalize that improving governance is essentially SDG work, given the robust link between WGI scores and development indicators .

One immediate implication is the need to prioritize the lagging SDGs identified by the study (notably SDG 6 in the case of Morocco, and other country-specific weaknesses) in national development plans. In Morocco, for example, top officials should urgently address the water and sanitation crisis by overhauling water governance: this could involve investing in climate-resilient water infrastructure, tightening regulation of water usage, and instituting better accountability mechanisms for service delivery. Across the region, governments should similarly target their most problematic SDGs with governance-centric interventions. If inequality (SDG 10) is a weak point, as in parts of Algeria and Egypt, then policies might include reforming subsidy systems, ensuring impartial welfare distribution, and improving public financial management to reach vulnerable communities. If hunger (SDG 2) persists, as indicated for Algeria and Tunisia, agricultural governance and rural development programs must be strengthened to deliver inclusive growth. The grey relational framework from this study can serve as a decision-making tool: policymakers can use it to pinpoint which SDGs are least responsive and trace those back to specific governance deficits. A critical managerial implication is the emphasis on monitoring and accountability in the SDG implementation process. North African governments are encouraged to improve their SDG monitoring systems by incorporating governance metrics. For instance, regularly tracking WGI dimensions alongside SDG indicators can help flag issues early such as a decline in government effectiveness or rule of law that might be undermining progress in a particular goal. Institutional leaders (in planning commissions or SDG oversight bodies) should establish clear links between governance reforms and SDG targets, creating accountability loops. One concrete step could be to tie budget allocations or donor support to governance outcomes in key sectors: e.g., extra funding for water and health projects could be conditioned on transparent procurement and citizen oversight, to ensure money is well-spent. This fosters a culture where success is measured not only by kilometers of road built or schools opened, but also by improvements in governance quality such as reduced corruption in those projects. Such an approach echoes recommendations in the literature that African governments must address governance bottlenecks like corruption, lack of transparency, and weak institutions to achieve the 2030 Agenda goals .

In practice, this could mean implementing e-government systems to reduce bureaucratic corruption, strengthening anti-corruption agencies, and protecting media freedom to expose maladministration all managerial actions that bolster WGI scores and, by extension, SDG performance. North African leaders should also pursue peer learning and international best practices in governance. The study suggests looking at countries with high WGI scores to derive practical lessons. Policymakers can expand cooperation with international organizations and governance initiatives (such as the African Peer Review Mechanism or UNDP's governance programs) to learn how improved voice and accountability or regulatory quality have translated into development gains elsewhere. For example, Tunisia and Morocco might share insights on successful initiatives in renewable energy (SDG 7) and institutional reforms, while learning from countries like Rwanda or Senegal on community health governance. At the managerial level, adopting proven tools like transparent public dashboards for SDG progress or participatory budgeting processes can bridge the gap between high-level commitments and on-the-ground results. In essence, North African governments must manage the SDG process not just as a set of technical targets but as a governance challenge: this entails fostering inclusive decision-making, evidence-based policy adjustments, and long-term institutional capacity. By doing so, leaders will be better equipped to navigate trade-offs and synergies among goals. For instance, managing water scarcity (SDG 6) in Morocco requires coordination across agriculture (SDG 2), climate (SDG 13), and industry (SDG 9) a whole-of-government approach that only effective governance can deliver. Ultimately, the managerial takeaway is that sustainable development in North Africa will advance fastest where governance reforms and SDG actions go hand in hand. As one recent analysis succinctly put it, governance and sustainable development are intrinsically intertwined, implying that ministries of finance, environment, education, etc., all need to integrate governance improvements into their sectoral strategies for the SDGs.

6.2. Practical and social implications

Beyond the corridors of government, the study's findings also carry broader practical and social implications for civil society, the private sector, and international development partners in North Africa. One key implication is the vital role of civil society and citizen engagement in achieving sustainable development. The strong link observed between voice & accountability and SDG outcomes highlights that when citizens can freely express demands and hold authorities accountable, development programs tend to be more responsive and equitable. Therefore, civil society organizations (CSOs), community groups, and the media in North Africa should leverage this insight to advocate for greater transparency and public participation in SDG-related initiatives. For example, local NGOs working on water and sanitation can organize community monitoring of water projects, giving feedback to authorities and helping ensure that resources earmarked for SDG 6 reach the intended beneficiaries. Grassroots movements such as youth-led climate activism or women's rights campaigns can also use the evidence of this study to press for policy changes, knowing that closing governance gaps (like gender inclusion in decision-making) can directly improve outcomes on goals like SDG 5 and SDG 10. In contexts where governments are less open, social actors may need support and protection; hence, another implication is the need for legal and institutional safeguards for civic space. Ensuring that citizens can voice concerns without repression will enable society to contribute to and watchdog the SDG process. This is especially pertinent considering that freedom of expression and democratic participation remain limited in parts of the region, and the study's conclusions explicitly call for action to enhance these freedoms to boost governance quality.

The private sector and non-governmental organizations also have practical steps to take. North African businesses and industry leaders should recognize their stake in governance and sustainable development, as emphasized by the study's integrated framework. Sustainable development is not solely the government's domain; companies can be both beneficiaries of good governance (e.g. fair regulations, stable environments) and contributors to SDG progress. A practical implication is that firms should align their strategies with SDG targets and incorporate governance considerations into their operations. For instance, a corporation in Morocco in the agriculture or mining sector could invest in water-efficient technologies and publicly report its water usage, thereby contributing to SDG 6 while exemplifying transparency and accountability. The research provides a framework for organizations to assess the long-term impacts of their activities on society and track their contributions to the SDGs.

Adopting such a framework, perhaps akin to sustainability reporting or ESG (Environmental, Social, Governance) standards, will encourage businesses to plan beyond short-term profits and consider broader societal well-being. This kind of voluntary accountability can generate valuable data and insights for policymakers (as noted in the study), essentially creating a feedback loop where private initiatives inform public strategy. Moreover, public-private partnerships can emerge from this collaborative stance for example, tech companies and governments partnering to improve digital governance systems for education (SDG 4) or startups working with cities to deploy smart water management solutions (SDG 11 and SDG 6). The practical implication is clear: sustainable development in North Africa requires an all-hands approach, where the private sector and civil society work in tandem with governments to address governance shortcomings. By doing so, they not only mitigate risks (like social unrest or resource crises) that can hamper business and community life, but also actively help fill capacity gaps in delivering services. International organizations and the broader global community also have a role, underscoring social implications that transcend national borders. North African countries' struggles with certain SDGs especially those linked to climate change, water scarcity, and inequality have global resonance. International development partners (UN agencies, the African Union, World Bank, etc.) should draw on the study's insights to tailor their support. Rather than funding siloed projects, aid programs should incorporate governance capacity-building components, recognizing that a well-governed sector will sustain results longer. For example, if a development bank finances a clean water project in Algeria or Sudan, it should also invest in training local water authorities, establishing anti-corruption safeguards, and involving local communities in oversight. This approach ensures that infrastructure and institutions grow together. The evidence that poor governance in areas like water and food security can heighten social instability and even conflict risk sends a strong signal to international actors: helping North Africa achieve SDG goals is also an investment in peace and regional security. Thus, diplomatic and aid efforts might focus on facilitating dialogue and agreements on transboundary water management (given that rivers and aquifers often span borders in this region) or supporting platforms for civic engagement and youth empowerment across the Maghreb. The social implication is that improving governance and SDG outcomes in North Africa will require solidarity and knowledge-sharing across society – from the local village council up to multilateral institutions. In conclusion, the governance–SDG nexus illuminated by this study implies that everyone in society has a part to play. Citizens and civil society are called upon to participate, advocate, and hold leaders accountable, ensuring that governance reforms do not stall, and that development truly meets people's needs. Policymakers and managers must be responsive to this input and willing to make difficult reforms in how institutions operate. The private sector is encouraged to become a proactive partner in sustainable development, adopting responsible practices that complement public policy. And international partners should reinforce these efforts by providing both resources and a framework of principles (such as good governance norms and SDG benchmarks) to guide action. The social fabric of North Africa – its trust in institutions, the vibrancy of its civic life, and the engagement of its youth will ultimately determine how well governance improvements translate into sustainable development on the ground. The positive feedback loop is evident: better governance fosters development, and a more developed, educated society in turn demands and sustains better governance.

By capitalizing on this virtuous cycle, North African societies can tackle their most pressing challenges, from water scarcity to job creation, and move closer to the inclusive and sustainable future envisioned by the SDGs.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT authorship contribution statement

Muhammad Ikram: Writing – original draft, Validation, Software, Project administration, Investigation, Data curation, Writing – review & editing, Visualization, Supervision, Resources, Methodology, Formal analysis, Conceptualization. **Chaymae Boudraa:** Visualization, Conceptualization, Writing – original draft, Validation.

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